



ASPIRA

**supervised by
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ASPIRA

IT 496: Graduation Project Report Product Release-1

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second Semester -1446
Fall 2024-2025



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chapter 1: Introduction



1. Introduction

The transition from education to professional life is a crucial phase that defines an individual's career path. In Saudi Arabia, this challenge has become a key focus as the country works towards Vision 2030, emphasizing the empowerment of youth by bridging the gap between academic education and professional development. However, many students and graduates face challenges during this transition due to the lack of personalized career guidance and targeted skill-building opportunities.

This project addresses the issue of insufficient mentorship and career guidance available to students and graduates. Without proper support, many struggle to enter the workforce with confidence, leading to a skills gap that impacts their ability to effectively participate in the job market.

To tackle this, we propose the development of a web-based platform called ASPIRA, designed to provide personalized mentorship and career development support through technology. By leveraging artificial intelligence, the platform aims to enhance accessibility and offer tailored guidance through features such as AI-powered chatbot, recommendation system, and a unified hub connecting mentees with professional mentors. This platform aims to equip students and graduates with the skills, confidence, and direction they need to succeed in their careers.

This document will provide a detailed description of the problem being addressed, outline the proposed solution, and include details about the technologies used, methodologies adopted, and expected outcomes of the project, aligning the initiative with the objectives of our Vision.



1.1. The Problem

In Saudi Arabia, a significant real-world problem persists: many mentees, particularly bachelor's students and recent graduates, struggle to transition effectively from education to the workforce. This challenge is particularly critical because it affects their ability to contribute meaningfully to the country's growth and long-term objectives.

For example, consider a university graduate who, despite excelling academically, finds it difficult to identify the right career path or acquire the skills needed for specific roles in the competitive job market. The lack of accessible and personalized career guidance often leaves individuals like this graduate unprepared to navigate the complexities of professional development.

Our project focuses on addressing a critical part of this issue: the absence of tailored mentorship and skill-building opportunities. Many mentees lack regular access to professional mentors who can provide career advice tailored to their goals and interests. This gap hinders their ability to plan effectively for their careers, build confidence, and acquire relevant skills. By bridging this gap, we aim to empower youth to achieve their professional potential and align with national aspirations for a skilled and competitive workforce.



1.2. The Solution

This project is designed to revolutionize career mentorship by creating a smart, AI-powered platform that bridges the gap between mentees and professional mentors. The goal is to provide personalized guidance, structured mentorship, and tailored learning resources, helping mentees navigate their career paths with confidence. Unlike traditional systems where finding the right mentor can be difficult and guidance is often generalized, this platform ensures that mentees are matched with mentors who align with their interests, career aspirations, and expertise needs.

The platform's Mentor Management system, handled by an admin, ensures that only qualified and verified mentors are approved, maintaining a high standard of mentorship. To make mentorship more accessible and effective, mentees can search for and filter mentors based on criteria such as industry and years of experience, ensuring a personalized experience. Once connected, mentees can book private one-on-one mentorship sessions or participate in interactive group sessions through the Zegocloud API [1], enabling seamless video communication for career discussions, skill-building, and guidance. Mentees can reschedule or cancel their bookings, while mentors have the ability to adjust or remove sessions as needed. To ensure clear communication, mentees will receive a notification whenever a mentor reschedules or cancels a booked private session. This keeps them informed about any changes and allows them to manage their schedules effectively. To enhance the learning experience, each session is documented, with mentors responsible for writing summaries so mentees can review key takeaways at any time.

Beyond mentorship sessions, the platform actively supports mentees' career growth through an AI-powered recommendation system that suggests relevant courses based on user interests, ensuring they develop skills aligned with their professional goals. Additionally, an AI-powered chatbot, integrated using OpenAI's [2] technology, provides instant assistance with platform navigation, session management, and career-related inquiries, ensuring mentees always have access to the information they need.

Mentees and mentors have full control over their profiles, allowing them to update their information, as their careers progress. To further support mentees in their professional growth, the platform includes a CV Builder, which helps mentees create a professional CV by entering



their personal details, education, skills, and work experience. This feature allows mentees to generate CV for job applications and networking opportunities.

By combining AI-driven learning, structured mentorship, real-time interactions, and a strong community-driven approach, this platform provides an essential resource for career development. It directly supports Vision 2030 by equipping Saudi youth with the skills, mentorship, and guidance they need to thrive in a competitive job market. By fostering a culture of mentorship, skill-building, and knowledge-sharing, the platform empowers a new generation of professionals, contributing to national growth, economic innovation, and workforce readiness.

1.3. Product

1.3.1. Product Vision

For mentees who are bachelor's students and fresh graduates.

Who need personalized career guidance and skill-building opportunities to transition into the workforce.

The ASPIRA platform **is** a web-based career development and mentorship platform .

That provides tailored real-time mentorship.

Unlike traditional mentorship programs or generic online mentoring platforms.

Our product uses AI to offer personalized guidance, a dynamic recommendation system for a seamless user experience, and AI-powered support, a chatbot to help users achieve their career goals.

1.3.2. Product Roadmap

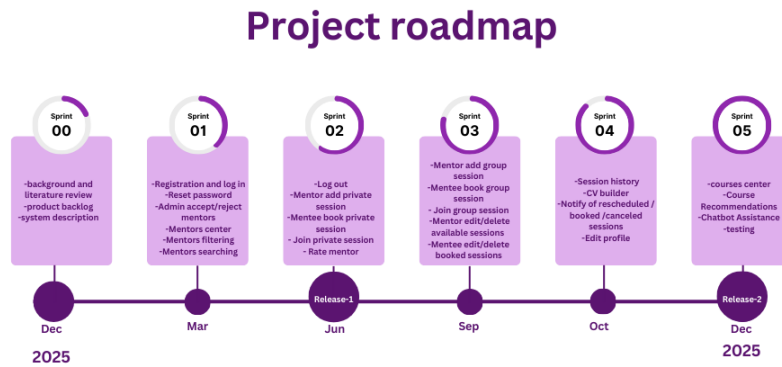


Figure 1 Product Roadmap

1.3.3. Objectives

Our objective is to develop the "ASPIRA" career guidance platform, designed to bridge the gap between mentees and mentors across various fields. This platform will provide several key features, including:

- **Product (customer focus-value):**
 1. **Mentors / Mentee Registration:** Enable mentors and mentees to register to access the platform's features, with dedicated interfaces for each user type.
 2. **User (Mentor/Mentee/Admin) Login:** Enable users to log in their accounts to access the platform.
 3. **Mentor Management:** Enable Admin to view and manage the list of all registered mentors and approve or reject mentors based on their provided qualifications and information.
 4. **CV Builder:** Mentees can enter their personal details, education, skills and work experience to create a professional CV for career growth opportunities.



5. **Mentors center:** A dedicated center displaying mentors profiles with information such as their name, photo, experience, availability, and brief description, enabling mentees to search mentors.
6. **Mentor Filtering:** Enable mentees to filter mentors based on their field of expertise, Experience years and ensuring a personalized and efficient search experience.
7. **Mentor searching:** Enable mentees to search for mentors using specific criteria, such as name, to easily find the mentor that they are looking for.
8. **Private Session Booking:** Enable mentees to easily book private mentorship sessions.
9. **Join private sessions:** Utilizes Zegocloud API, that has embedded video call functionality, to enable mentees to join live video private sessions.
10. **Group Sessions:** to enable mentees to easily book group mentorship sessions.
11. **Join group sessions:** Utilize Zegocloud API, to allow mentees to join live video calls, for general career guidance and advice, fostering a collaborative and inclusive learning environment.
12. **Mentor Rating:** Enables mentees to rate their sessions with mentors to improve platform reliability.
13. **Past sessions:** Contains a list for both mentors and mentees previous sessions, including date/time and name. It will also include a session summary for the mentees prepared by the mentor, providing a summary of each session, making it easy for users to refer back to the main points discussed. Also mentees can rate each session, providing feedback on the mentor's performance through a star rating and a comment.
14. **Upcoming session:** Keep track of future sessions for both mentors and mentees.
15. **Add Sessions:** The mentors can add and display sessions available to mentees. Mentees can choose and book from this list of available sessions.
16. **Mentee session management:** Mentee can reschedule only private sessions to a different available time slot or delete the booking of private/group sessions if they no longer need the session.



17. **Mentor session management:** Mentor can edit the time of private and group sessions or delete scheduled sessions if needed.
18. **courses center:** Dedicated center displaying courses information such as their name and online link, providing an easy way for mentees to explore available learning opportunities.
19. **Mentee Notifications:** Mentees will be notified when the mentor reschedules or deletes the booked private/group sessions, this ensures they stay informed about any changes.
20. **Mentor Notifications:** Mentors will be notified when a mentee book or reschedules or delete a private session, to keep up with any changes.
21. **Course Recommendations:** Implement an AI recommendation system specifically designed to suggest relevant course content tailored to user interests. The system leverages user input to provide personalized recommendations, ensuring users find the most suitable and engaging course materials for their needs.
22. **Chat Assistance:** Integrate an OpenAI-powered chatbot to assist users with platform navigation and career guidance, providing instant support and recommendations.
23. **Profile Management:** Enable mentors and mentees to edit their profiles.
24. **Intuitive Interface:** Ensure a user-friendly experience for seamless navigation, session booking, and resource access.



- **Project (solution focus-plan):**

1. **User Research:** Conduct surveys and interviews with mentees, and mentors to identify challenges and expectations.
2. **Requirement Analysis:** Define functional and non-functional requirements to ensure user-centric platform features.
3. **Background research and literature review:** conduct domain analysis along with a comprehensive competitive analysis
4. **System Design:** Create architecture and wireframes for an intuitive and interactive platform interface.
5. **Data Collection:** Prepare mentor profiles, FAQs, and course data for integration.
6. **Feature Development:** Build key features, including the chatbot, recommendation system, and session booking functionality.
7. **Testing and Feedback:** Conduct usability testing, gather user feedback, and refine the platform for reliability and ease of use.
8. **Launch and Monitoring:** Deploy the platform and monitor performance for continuous improvement.

- **Learning (student focus):**

By completing the ASPIRA project, we aim to learn new skills that will help us in real-world software development, including:

1. **Frontend Development:** Design a responsive and user-friendly interface.
2. **Backend Development:** Build robust data management and processing systems.
3. **Database Management:** Design and manage efficient databases to handle user profiles, scheduling, and platform data.
4. **AI Integration:** Implement content-based recommendation systems and chatbot functionality.
5. **System Integration:** Connect frontend, backend, and databases for a seamless platform experience.
6. **Team Collaboration:** Improve teamwork and communication skills.

1.3.4. Scope

ASPIRA is a mentorship platform designed to connect mentees with experienced mentors who align with their career goals, interests, and fields of study. It provides a structured and interactive environment where mentees can seek guidance, gain industry insights, and receive personalized support throughout their professional journey. Unlike traditional mentorship programs, ASPIRA leverages technology to ensure accessibility, efficiency, and engagement.

ASPIRA offers private one-on-one mentorship sessions and interactive group sessions, allowing mentees to connect with mentors in real-time. Users (mentor and mentees) can easily reschedule or cancel sessions, while the notify mentee feature ensures that mentees receive updates when a mentor reschedules or cancels a booked private session. To enhance mentorship impact, mentors are responsible to summarize each session, for mentees to revisit for future reference. Additionally, mentees can rate mentors after sessions, helping maintain quality guidance. Furthermore, ASPIRA is fully accessible across all devices, ensuring seamless interaction anytime, anywhere. ASPIRA simplifies mentor discovery through its Mentor Center, where mentees can filter mentors by field or search for specific names. Besides the CV Builder feature allows mentees to input their details and expertise to create a CV. To enhance the user experience, ASPIRA integrates AI-driven features, an OpenAI-powered chatbot that provides instant support for navigation, mentorship queries, and career advice. Additionally, a Course Center with a recommendation system to suggest relevant learning opportunities based on mentees' interests and career objectives. ASPIRA is a mentorship and career guidance platform, not an educational course provider. While it offers intelligent course recommendations, it does not handle course registration, enrollment, or payment processing. Its primary mission is to foster meaningful mentor-mentee connections, support career growth, and provide structured, AI-enhanced guidance.



1.3.5. Hardware/Software Tools and Cost

Table 1 Hardware/Software Tools and Cost table

Hardware Tools	
Name and Description	Cost
Laptop	Every student has their own device.
Software Tools	
Name and Description	Cost
MySQL (Database Management) [3]	Free
Visual Studio code is A simple code editor that helps with tasks like debugging, running jobs, and keeping track of different versions of your code and supports almost every major programming language." [4]	Free



course provider API a simple service that provides access to a list of courses, allowing users to browse and view detailed course information effortlessly.	To be defined
GitHub is a web-based hosting service for git repositories. [5]	Free
Jira "It's the agile project management tool that teams use to plan, track, release, and support high-quality software with confidence." [6]	Free
ZeGOcloud API A simple service that provides voice, video, and live streaming capabilities, allowing users to easily integrate communication features into their websites or applications. [1]	Depends on the number of calls, one call costs \$0.99
Open ai API A simple service that provides access to advanced AI models, allowing users to easily integrate features like text generation and conversation into their applications and websites. [2]	Pricing depends on subscription and usage. For example, ChatGPT 4 costs about \$0.0015 per 1,000 input tokens and \$0.002 per 1,000 output tokens.



1.4. Scrum Team

1.4.1. Skill Set Requirements

Table 2 Skill Set Requirements table

Technical Skill Required	What is the current level of the team (beginner-intermediate- advanced) for each skill? How will the gap be bridged? (if necessary) Learning plan
Programming languages	Intermediate, additional self-learning and online courses.
Dealing with APIs	Beginner (We can enhance our API skills by reviewing fundamental concepts such as API structure from the specification document).
Database Management (MySQL)	Intermediate, Follow online tutorials and build database-driven applications
Training the AI models	Beginner (We can strengthen our skills by addressing the fundamental concept behind the AI model and seeking for small projects).
UI-UX skills	Advanced



1.4.1.1. Learning

In this section, we will discuss how we acquired the necessary knowledge. To build an effective ASPIRA, we explored various technologies, methodologies, and best practices relevant to career advising, chatbot development, recommendation systems, and video session management.

To build ASPIRA, we learned about career advising by researching different mentorship models and best practices.

For chatbot development, we watched video tutorials on natural language processing (NLP). We also explored OpenAI's chatbot models and read about their ability to provide career advice.

To implement the recommendation system, we followed online tutorials on content-based and API integration. We practiced retrieving and structuring data to suggest relevant learning resources for mentees.

Additionally, we explored video sessions that explain details about API to manage user interactions within the platform.



1.4.2. Roles and Responsibilities

Table 3 Roles and Responsibilities table

Scrum Team	
Product Owner:	Dr. Arwa Alsultan
Developers:	Aeshah Almuklifi Luluh Alyahya Shahad Alsalman Ghadi Alzahrani Noura Alwohaibi
Scrum Master (SM):	Dr. Arwa Alsultan
Stakeholders:	Examiners and mentees



chapter 2: Background



2. Background

In this section, we explore career advising and its role in helping individuals make informed decisions about their professional paths. Section 2.1 introduces career advising, discussing its importance in guiding mentees through career choices, skill development, and job market preparation. It also covers different approaches to career advising, including directive, assessment-based, and specialized methods, each addressing unique challenges mentees may face. Additionally, this section explores the roles of mentors and mentees and the impact of structured guidance. It also highlights how technology is revolutionizing career advising, making mentorship more accessible and tailored to individual needs.

Section 2.2 introduces the impact of chatbots in modern mentorship platforms. With AI transforming the way we seek advice, chatbots are now capable of providing real-time career guidance, answering mentees' questions, and even offering personalized suggestions. We specifically look at OpenAI-powered chatbots and how they enhance mentorship by making conversations more interactive, informative, and relevant.

Next, in Section 2.3, we explore how recommendation systems help mentees find the right learning opportunities. ASPIRA's recommendation system is designed to provide highly personalized career guidance, ensuring that mentees receive meaningful suggestions tailored to their aspirations.

This background lays the foundation for ASPIRA, a mentorship platform that leverages career advising, chatbots, and recommendation systems to create a smarter and more accessible mentorship experience. By integrating these technologies, ASPIRA helps mentees make confident decisions and take the right steps toward their professional goals.



2.1 Career Advising

Career advice plays a vital role in both personal and professional development, providing mentees with the support and guidance they need to navigate their education and career choices. It's a collaborative process where mentors help mentees discover their strengths, set realistic goals, and make informed decisions about their future. Through structured guidance, career advising enables mentees to explore different career options, assess their abilities, and develop the skills necessary to excel in their chosen fields. Career mentors play a crucial role in this journey, helping mentees identify their skills and interests to ensure they choose career paths that align with their capabilities and aspirations. [7]

This background discussion will explore the core aspects of career advice, how it helps mentees unlock their potential, and its impact on making well-informed career decisions throughout their professional journeys.

2.1.1 Career Advising Definition and Significance

Career advice serves an essential role in helping mentees navigate their professional journeys. At its core, career advice is a process of guidance and mentorship aimed at empowering mentees to make informed decisions about their future. It encompasses a range of activities, from identifying strengths and setting achievable goals to exploring opportunities and overcoming challenges.

The significance of career advice lies in its ability to provide clarity and direction during important stages of a person's life. Whether someone is choosing a field of study, entering the job market, or considering a career change, sound advice can make all the difference. It helps mentees understand their unique skills, interests, and potential while aligning these with market demands and opportunities.[8]

Career advice is a flexible and tailored approach focused on building a strong relationship between mentors and mentees. This approach creates a supportive environment where mentees can reflect on their aspirations, evaluate their options, and develop strategies for success. By offering insights into career trends and industry expectations, mentors help mentees navigate the complexities of the job market and align their decisions with long-term goals. [9]



In addition to providing guidance, career advice equips mentees with practical tools such as crafting resumes, preparing for interviews, exploring career transitions, and understanding industry-specific trends. It also delivers actionable strategies to address career-related challenges, from managing competition to building professional networks. By combining personalized mentorship with practical support, career advice empowers mentees to confidently pursue their aspirations in their professional journeys.

Also career advice fosters personal growth. It helps mentees build confidence, adaptability, and problem-solving skills, which are essential in a constantly evolving job market. By offering guidance on time management, work-life balance, and professional development, career advice ensures that mentees are prepared to face challenges and achieve long-term success.[10]

2.1.2 Types of Career Advising and Challenges in Career Advising

Career advising includes various approaches designed to support mentees at different stages of their professional journey. The most common types include directive advising, assessment-based advising, and specialized advising, each offering unique methods to guide career decision-making with the help of mentors. However, despite the availability of these approaches, mentees often encounter significant challenges in navigating their career paths. Understanding both the types of career advising and the obstacles mentees face is crucial in providing effective guidance and support.

- Directive Advising:

Directive advising is a hands-on approach where mentors provide clear, specific guidance to help mentees make confident career decisions. Rather than exploring multiple options, mentors offer direct solutions and practical strategies for tasks such as choosing between job offers, improving resumes, or preparing for interviews. This approach is particularly beneficial for mentees who need quick answers or prefer straightforward advice from a mentor. [11] However, many mentees face challenges that hinder their career progress, such as uncertainty about their career direction and a lack of personalized guidance. Where many struggle to identify which career path suits them best, what opportunities align with their skills, or how to



take the next step, leaving them feeling lost. Additionally, generic career advice often fails to consider a mentee's unique strengths, interests, and aspirations, making it difficult to apply broad recommendations to their specific situations.[12]

- **Assessment-Based Advising:**

relies on personality tests, aptitude assessments, and skills evaluations to help mentees understand their strengths and align them with suitable career paths. This data-driven approach provides valuable insights into professions that match their abilities and interests, guiding them toward informed career decisions.[11] However, mentees often face challenges that complicate career planning, such as skills gap issues and the rapidly changing job market. Many job seekers, especially graduates, struggle with misalignment between their skills and employer expectations, making it difficult to secure job opportunities or advance professionally. Additionally, as industries evolve with new technologies and shifting demands, mentees must continuously assess their competencies and determine whether further training or education is needed to remain competitive.[12]

- **Specialized Career Advising:**

Specialized career advising provides a personalized approach for mentees facing unique career challenges. Specialized mentors support students exploring career paths, young professionals seeking growth, career changers navigating transitions, and individuals overcoming obstacles such as workplace discrimination or skill gaps. By understanding each mentee's specific situation, these mentors offer tailored guidance to help them progress in their careers. [11]However, many mentees struggle with self-doubt and fear of failure, especially during career transitions and workplace challenges, making it difficult to take proactive steps toward their professional goals. Specialized career advising addresses these concerns by offering targeted strategies for overcoming workplace obstacles, acquiring new skills, and successfully navigating career changes. Through personalized support, mentees gain the confidence and resources needed to advance in their careers with clarity and determination.[12]



Career advising helps mentees navigate their careers through different approaches, from direct guidance to skill assessments and personalized support. Despite these resources, challenges like career uncertainty, skill gaps, and self-doubt and fear of failure. By understanding these obstacles and tailoring advising methods, mentees can receive the right support to make informed decisions and confidently pursue their goals.

2.1.3 The Role of Mentors and Mentees in Career Advising

Mentors and mentees both have an important role in career advising, creating a relationship built on guidance, learning, and professional growth. Mentors use their experience and industry knowledge to help mentees navigate career choices, develop skills, and build confidence. They offer advice, share real-world insights, and provide support in setting career goals. But a mentor's role isn't just about giving answers it's about listening, encouraging mentees to think for themselves, and guiding them to make smart career choices.[13]

At the same time, mentees need to take an active role in their career growth. They should seek guidance, ask relevant questions, and use the advice they receive. A good mentoring relationship works best when mentees are open to feedback, eager to learn, and take initiative in building their careers.[13]

When both mentors and mentees are actively involved, career advising becomes a valuable tool for growth, networking, and long-term success. It's more than just answering career questions—it helps build confidence, open new opportunities, and provide a clear path for the future.

2.1.4 The Role of Technology in Career Advising

Technology has transformed career advising, making it more accessible and efficient than ever before. Digital tools now offer real-time guidance, personalized career suggestions, and interactive learning, making it easier for mentees to explore career options and build their skills.

One of the biggest advantages of technology in career advising is its ability to offer personalized guidance. AI can analyze a mentee's skills, interests, and job market trends to



recommend career options that best match their strengths. Online assessment tools, such as personality tests and skills evaluations, give mentees a deeper understanding of their abilities, helping them make career choices and explore opportunities that align with their goals.

Technology also plays a crucial role in skill development, making learning more accessible and flexible. Online platforms offer a wide range of courses, certifications, and training programs that help individuals build the skills needed for career growth. Whether it's enhancing technical abilities, staying updated with industry trends, or improving soft skills, these digital resources allow mentees to learn and upskill at their own pace. Career advising is no longer just about offering guidance it now includes access to tools that actively support skill-building and career readiness.[14]

Another major shift in career advising is how technology has transformed mentorship and networking. Virtual platforms now allow mentees to connect with mentors from anywhere, giving them access to industry experts and career coaches without geographic limitations. Video calls, webinars, and online career communities have made it easier than ever to seek guidance and build professional relationships.

This transformation in technology and career advising inspired us to create ASPIRA. By leveraging these advancements, we developed a platform that offers personalized career guidance, skill-building opportunities, and direct mentorship connections. ASPIRA provides a space where mentees can connect with mentors who can guide them through career challenges and opportunities. Our goal is to make career advising more accessible and empowering mentees to confidently take the next steps in their professional journey.

Overall, technology has made career advising easier and more accessible. It helps mentees take charge of their careers, get expert advice anytime, and make smart decisions based on real-time job market insights. As technology advances, career advising will become even more interactive.

2.2 Chatbots: An overview

Chatbots have become an integral part of our everyday interactions with technology, fundamentally changing how we communicate with systems and access information. These



virtual assistants are designed to simulate human-like conversations, making communication feel intuitive and natural. At the core of chatbots lies Natural Language Processing (NLP), a branch of artificial intelligence that enables them to understand, process, and respond to human language in meaningful ways. Over the years, chatbots have evolved significantly, transitioning from basic systems with limited capabilities to advanced tools that can engage in complex, context-aware interactions.

There are three primary types of chatbots, each suited to different needs. First, Rule-Based Chatbots operate based on predefined rules or scripts. These are the simplest form of chatbots, following decision trees to guide users through fixed workflows. They work well for straightforward tasks like answering frequently asked questions or providing basic information but struggle when users deviate from their programmed paths. Second, AI-Powered Chatbots take things to the next level. These bots use machine learning and natural language understanding to interpret user input, learn from interactions, and adapt to context. They are capable of dynamic conversations, making them ideal for handling more complex queries, providing personalized recommendations, or even assisting with problem-solving. Finally, Hybrid Chatbots combine the best of both worlds, using rule-based logic for simple tasks while leveraging AI for more intricate or unpredictable scenarios. This balance allows them to handle a broader range of interactions while maintaining efficiency.

Despite their growing capabilities, chatbots come with both strengths and challenges. Among their most significant advantages is their ability to provide 24/7 support, ensuring users receive help at any time without delays. They also excel at scalability, managing multiple conversations simultaneously, which is invaluable for businesses looking to handle high volumes of queries. Moreover, their ability to personalize interactions based on user data creates a more engaging and relevant experience for individuals. This is particularly useful in industries like e-commerce, customer service, and healthcare, where personalization can significantly enhance satisfaction and loyalty. Additionally, chatbots can help businesses save costs by automating repetitive tasks and reducing the need for human intervention in routine inquiries.

However, like any technology, chatbots are not without their challenges. Rule-based chatbots, while efficient for simple tasks, often fail to handle more nuanced or open-ended



queries, leading to user frustration. For more advanced, AI-powered chatbots, the development process can be highly complex, requiring expertise in artificial intelligence, machine learning, and natural language processing. Another significant concern is privacy and data security, particularly when chatbots handle sensitive information, such as in healthcare or finance. Striking the right balance between personalization and protecting user data is an ongoing challenge for developers and businesses alike.

2.2.1 OpenAI Chatbots

Natural Language Processing (NLP) has transitioned from relying on task-specific representations and architectures to adopting task-agnostic pre-training and architectures. This shift has significantly advanced various complex NLP tasks, including reading comprehension, question answering, and textual entailment. Although the foundational architecture and initial representations are now task-agnostic, a final task-specific adaptation step remains necessary. This involves fine-tuning the pre-trained model on a large dataset of examples, ensuring it is optimized for a specific task. [15]

One of the standout features of OpenAI chatbots is their versatility. They can handle both open-domain and closed-domain queries. For example, they can answer a broad range of questions on virtually any topic when used in an open-domain context, similar to ChatGPT. At the same time, they can be fine-tuned to operate within a specific domain, like mentoring or academic advising, to provide more focused and relevant responses. This flexibility allows OpenAI chatbots to be deployed across diverse scenarios, making them suitable for everything from casual conversation to specialized applications. Furthermore, their ability to continuously learn and improve ensures that they remain relevant and effective over time, adapting to new information and user needs with ease. When designing ASPIRA, our goal was to create a mentorship platform that not only connected mentees and mentors but also offered intelligent, meaningful, and personalized support throughout the mentorship journey. To achieve this, we needed a chatbot solution that could match the vision and complexity of our platform. After evaluating several options, OpenAI's chatbot technology emerged as the most fitting choice for ASPIRA's needs.

GPT-4, short for Generative Pre-trained Transformer 4, is OpenAI's most advanced AI language model. It's designed to understand and generate human-like text with remarkable



accuracy, making it one of the most powerful tools for natural language processing. Compared to its predecessor, GPT-3.5, this model shows major improvements in reasoning, contextual understanding, and overall performance.

To put its capabilities to the test, GPT-4 was evaluated on a range of exams originally designed for humans—and the results were impressive. For instance, in a simulated bar exam, GPT-4 ranked in the top 10% of test takers, while GPT-3.5 placed in the bottom 10%. This leap in performance highlights the model's enhanced ability to handle complex reasoning and problem-solving tasks.

Beyond standardized tests, GPT-4 also outperformed previous AI models on key natural language processing (NLP) benchmarks. On the Massive Multitask Language Understanding (MMLU) test, which covers 57 subjects, it surpassed all prior models by a significant margin. Even when the test was translated into 26 different languages, GPT-4 outperformed the best English-language AI models in 24 of them, showing its strong multilingual capabilities.

What makes GPT-4 even more impressive is that it doesn't rely on benchmark-specific training or manual optimizations like many other state-of-the-art systems. It's simply better at understanding context, making connections, and generating relevant responses. Additionally, OpenAI has improved its safety measures, making the model more reliable and aligned with human intent.

In short, GPT-4 isn't just another AI upgrade—it's a major step forward in how machines understand and communicate with humans.[16]

One of the primary reasons we chose OpenAI's chatbot is its advanced natural language understanding. Mentorship often involves nuanced conversations about career planning, skill development, and personal growth. OpenAI's chatbot is equipped to handle such discussions with sensitivity and precision, ensuring that mentees and mentors feel heard and understood. Its ability to provide thoughtful, context-aware responses makes it an ideal companion for users seeking guidance, whether they need help setting goals, navigating challenges, or exploring new opportunities.

Additionally, OpenAI's technology is highly adaptable, allowing us to train the chatbot specifically for the mentorship domain. Unlike generic chatbots, ASPIRA's chatbot is fine-



tuned to address mentorship-specific queries, making it a valuable resource for mentees and mentors alike. For instance, if a mentee is unsure about which skills to develop for a particular career path, the chatbot can provide tailored advice based on industry trends and best practices. This domain-specific focus ensures that the chatbot adds meaningful value to the mentorship experience, acting as a reliable and knowledgeable assistant.

Another key factor in choosing OpenAI was its scalability and continuous improvement capabilities. As ASPIRA grows and evolves, so too can the chatbot. It can be updated with new data, insights, and features to stay aligned with the changing needs of our users. This means that the chatbot will not only remain relevant but will also improve over time, delivering an increasingly refined user experience.

Finally, OpenAI's track record of innovation and reliability gave us confidence that their chatbot technology would meet the high standards we envisioned for ASPIRA. By integrating OpenAI's chatbot, we have created a platform that goes beyond simply facilitating mentorship connections—it empowers users by providing intelligent, responsive, and personalized support. This decision reflects our commitment to leveraging cutting-edge technology to enhance the mentorship experience, ensuring that ASPIRA remains a trusted and transformative tool for personal and professional growth.

2.3 Recommendation systems

In today's world, where we are bombarded with endless choices—be it movies, courses, job listings, or products—recommendation systems have become essential. These smart systems help users find what they need by analyzing their preferences, behaviors, and past interactions. Whether it's Netflix suggesting your next binge-worthy show, an online learning platform recommending courses that match your career goals, or an e-commerce site curating a shopping list just for you, recommendation systems are shaping the way we interact with digital content.



2.3.1 How Do Recommendation Systems Work

At their core, recommendation systems use machine learning and statistical analysis to predict what a user might like based on various factors. They act like digital assistants, learning from your interactions to make increasingly accurate suggestions over time. The goal is to cut through the noise and present options that are truly relevant to you. [17]

2.3.2 Types of Recommendation Systems

Recommendation systems can be categorized into different types based on their underlying methodologies. Content-based filtering systems analyze item attributes and user preferences to suggest similar content, making them suitable for applications like movie recommendations. Collaborative filtering, on the other hand, relies on user behavior patterns, identifying groups with similar interests to generate suggestions. Hybrid systems integrate multiple approaches to enhance recommendation accuracy, often combining content-based and collaborative techniques. Finally, knowledge-based systems use predefined rules and expert knowledge to recommend items that match specific user needs, making them particularly effective in domains like job matching and personalized services. [18]

2.3.3 Applications of Recommendation Systems

Without recommendation systems, navigating the overwhelming number of choices available online would be exhausting. These systems make life easier by providing personalized experiences across different platforms:

- Streaming Services (like YouTube): They analyze your watching or listening habits to suggest content you'll likely enjoy.
- E-Commerce (Amazon, eBay): They track your browsing and purchasing history to recommend products that match your interests.
- Online Learning (Coursera, Udacity, edX): They suggest courses based on your skills, career goals, and past learning history.
- Job Portals (like LinkedIn): They match job seekers with relevant job listings based on their resumes, searches, and application history.



2.3.4 The Tech Behind Recommendation Systems

Recommendation systems use a combination of machine learning techniques to provide accurate suggestions tailored to users' preferences and needs. One key method is Natural Language Processing (NLP), which enables systems to analyze and interpret text-based data, such as course descriptions, job postings, and movie summaries, to make meaningful recommendations. Another important technique is TF-IDF (Term Frequency-Inverse Document Frequency), a mathematical approach that assesses the significance of words within a document. For example, in job recommendation systems, TF-IDF helps match keywords from a user's resume with those found in job descriptions, improving relevance. Additionally, Cosine Similarity plays a crucial role by measuring the similarity between two documents or sets of information. This technique is often used to compare a user's past interactions with new content, ensuring that recommendations align with their preferences. When combined, these techniques enhance the precision of recommendation systems by analyzing patterns and relationships within data.

With the rapid growth of digital information, users often face information overload, making it challenging to filter relevant content, particularly in news portals where articles are published massively. Readers need time to search for and read news, but the time relevance of articles diminishes quickly. To address this, recommendation systems play a crucial role in delivering personalized content efficiently. A notable study implemented a news recommendation system using the TF-IDF method, which assigns weights to each word in the news title and then applies cosine similarity to find similarities between stories. To evaluate the system's effectiveness, the researchers compared the recommended articles with readers' actual news history on the Microsoft News portal using a hit-rate metric, achieving an accuracy of 80.77%, indicating the strong relevance of the system's recommendations.

This demonstrates how TF-IDF and Cosine Similarity are effective in various domains, from job matching to personalized news recommendations, enhancing user experience by delivering highly relevant content.[19]

Recommendation systems offer significant business benefits by leveraging data filtering tools to drive revenue growth, enabling cross-selling and upselling through personalized recommendations. By presenting relevant suggestions, customers are more likely to discover



and purchase additional products, leading to increased sales. Additionally, these systems enhance customer satisfaction by tailoring recommendations to user preferences, making the platform feel more responsive and engaging, which fosters retention and a positive user experience. With personalization at scale, recommendation engines analyze customer data to create individualized profiles, increasing conversion rates by curating products specifically suited to each user. Furthermore, these systems generate informative reports that provide insights into customer behavior, preferences, and trends, allowing businesses to make data-driven decisions for continuous growth. They also reduce workload by automating recommendations, freeing employees from manually curating suggestions and enabling them to focus on other strategic tasks. Lastly, controlled retailing helps businesses showcase new or underexposed products, manage inventory effectively, and strategically influence purchasing behavior, ultimately driving targeted sales and optimizing retail performance.[20]

recommendation systems face several key challenges. One major challenge is fostering communities of interest while ensuring user privacy is respected. Balancing personalization with data protection remains a crucial concern. Another significant challenge is developing recommendation algorithms capable of integrating multiple data sources collected at different times. For newly registered users, personal information can be gathered during registration, while contextual data such as location and time can be inferred from their IP address. Additionally, recommendations are often influenced by items that have been frequently viewed, downloaded, or purchased by others, further shaping the system's ability to deliver relevant suggestions.[21]

While recommendation systems have significantly enhanced content discovery, they still face notable challenges. One major issue is the cold start problem, where new users or items lack sufficient interaction history, making accurate recommendations difficult. Additionally, data bias can arise if the training data is skewed, leading to recommendations that disproportionately favor certain types of content. Another challenge is over-personalization, where users are repeatedly shown content similar to their past interactions, limiting exposure to diverse recommendations and potentially reinforcing filter bubbles.



At ASPIRA, we've implemented a content-based recommendation system designed to guide mentees toward courses that align closely with their skills, interests, and career aspirations. By analyzing each mentee's unique data—such as their goals, expertise, and past interactions—the system delivers highly relevant recommendations, such as prioritizing data analytics courses for mentees interested in that field. What sets this system apart is its adaptability; it evolves with the mentee by factoring in feedback and progress, ensuring recommendations remain precise and aligned with their growth. This personalized and evolving approach empowers mentees to achieve their professional goals while demonstrating ASPIRA's commitment to leveraging intelligent technology for impactful mentorship.

At the end of the day, recommendation systems are more than just algorithms—they're the invisible hands shaping our digital experiences. As they continue to evolve, they'll become even better at delivering what we need, sometimes even before we realize we need it.



chapter 3: Literature review

3. Literature Review

The development of ASPIRA as a mentorship platform required performing a market analysis of existing mentorship and career development services. This review establishes framework strengths and weaknesses together with proposed areas of uniqueness for ASPIRA through its mentorship service and career development offerings. The analysis of current solutions compared with ASPIRA helps the team create an improved mentorship environment that produces tailored user experiences.

3.1 ASPIRA Competitive product analysis

1. HRDF Recruitment Services (Saudi Human Resource Development Fund)

The main priority of the HRDF program stems from government support to provide Saudi job seekers assistance from their academic phase until they secure employment. The organization provides Saudi Arabian job seekers with employment service along with vocational guidance and supports services and training services that align with the local employment market requirement.[22]

Features:

- Saudi job seekers benefit from recruitment programs "Tawfeer" and "Hadafe" because they match qualified candidates with suitable positions through these initiatives.
- The institution gives specialized assistance to students moving from studying to starting their professional careers.
- Workforce development materials match Saudi needs of the Saudi job market.

Limitations:

- The service lacks both academic counseling and specific academic goal guidance for young learners.
- The institution needs to enhance its student mentoring services by developing individualized development plans and personalized student support systems.

- The organization concentrates on matching candidates with employment opportunities but provides few supplementary career development services beyond basic placement services.

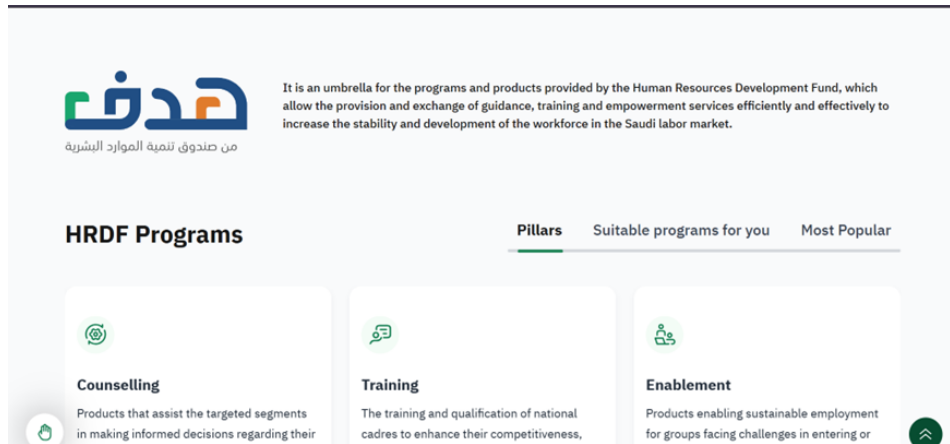


Figure 2 HDRF interface

2. MentorCity

The internet platform MentorCity helps students and working professionals search for mentors through their interests and educational goals and professional areas. The platform demonstrates both mentor-mentee relationship architecture and shows methods to set targets and check advancement levels.[23]

Features:

- Both mentor and mentee can find each other through this system based on matching their specific personal interests and career goals.
- Goal-setting features and tools for tracking progress.
- Flexible mentor-mentee matching system.

Limitations:

- Academic advising together with career services are not available through this system.
- Emphasis on professional mentorship, lacking comprehensive career development tools.

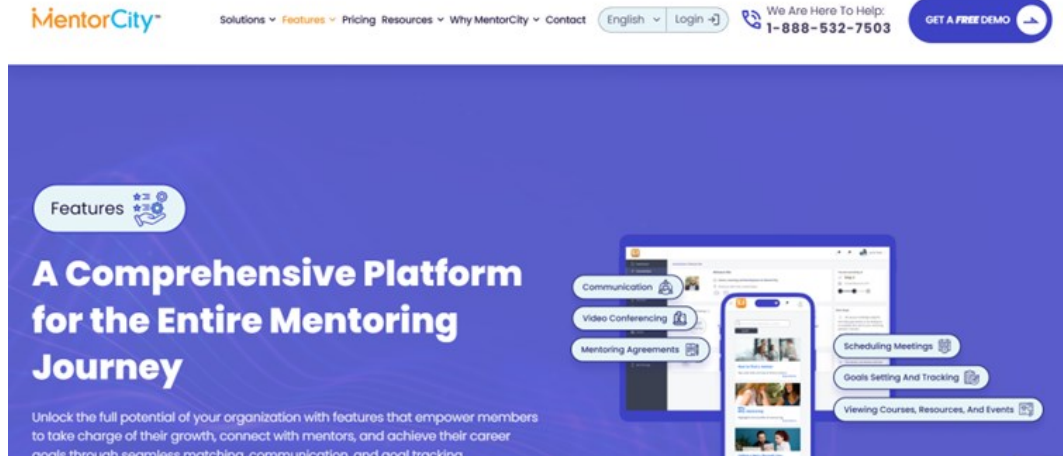


Figure 3 MentorCity interface

3. Alumni Networks (e.g., AlumniFire, Graduway)

The platforms AlumniFire together with Graduway offer networking functions between alumni members while simultaneously providing student-to-student interaction regarding mentorship and career advice connections. [24]

Features:

- The platform provides students the chance to network and seek advice from their university alumni.
- Career guidance from experienced professionals.
- Users create connections through their mutual employment fields and similar career paths as well as common work history.

Limitations:

- Primarily focused on alumni-student interactions.
- The service does not assist with education counseling or employment placement assistance.
- The platform does not offer custom-made suggestions for career guidance.



Figure 4 Graduway interface

4. LinkedIn Learning

A modern learning platform uses personal profile data for suggesting various courses to users. [25]

Features:

- Users benefit from extensive professional development material that is available for learning.
- Individual course suggestions exist for users whose profiles match their objectives.
- Users can integrate their LinkedIn profiles which shows their skills as well as displayed certifications.

Limitations:

- The program does not supply mentoring services or academic advising to its members.

- The platform provides skill training opportunities instead of employment support and career development coverage.

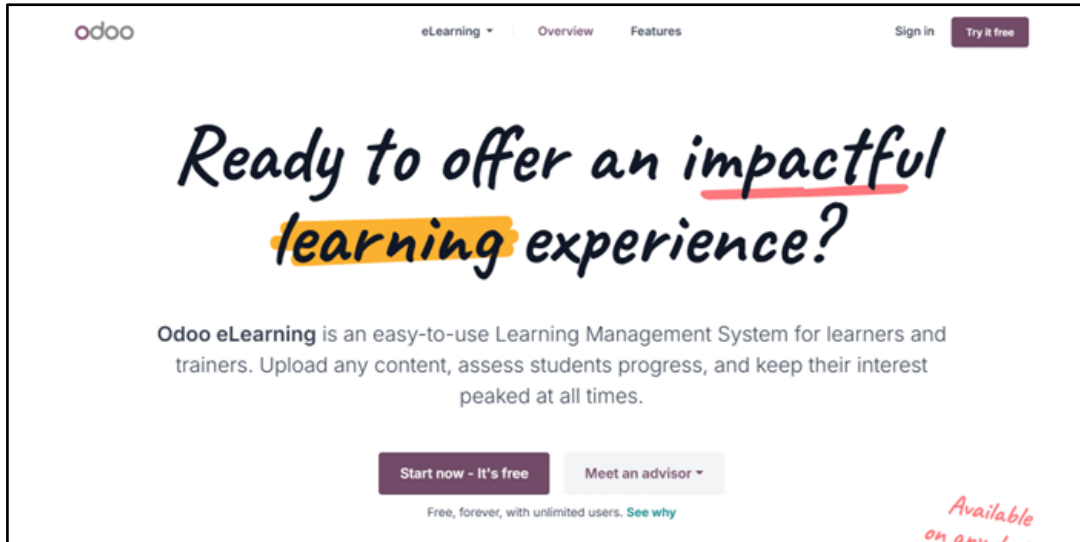


Figure 5 LinkedIn Learning interface

5. Qooper

Qooper serves as a platform which helps people find mentors through structured mentorship programs supported by automation and artificial intelligence recommendations.[26]

Features:

- AI-powered mentor-mentee matching.
- Local institutions exist which provide mentorship programs with automated goal tracking capabilities.
- The system includes automated warning features together with systems to monitor development progress.
- Integration with career development resources.

Limitations:

- This platform delivers corporate mentoring programs instead of academic counseling guidance.
- The system shows restrictions when students require individualized academic and career guidance.

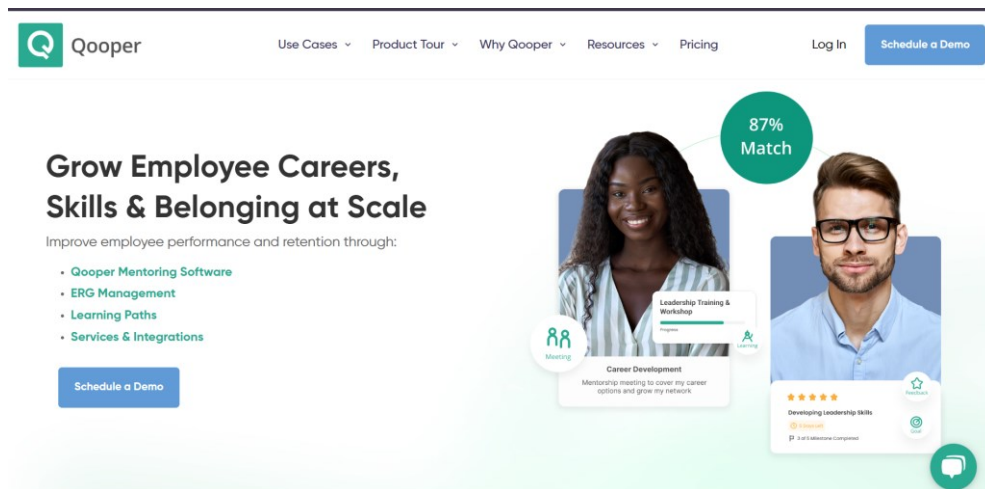


Figure 6 Qooper interface

6. MentorcliQ

Through automation MentorcliQ operates as a corporate mentoring platform that creates analytics reports and runs business mentoring programs. [27]

Features:

- AI-driven mentor matching.
- Customizable mentorship programs for organizations.
- Performance tracking and analytics for mentorship progress.
- Seamless integration with Learning Management Systems (LMS).

Limitations:

- The platform provides corporate mentoring services but has limited activity in student mentorship activities.
- Lacks academic guidance and career planning tools.

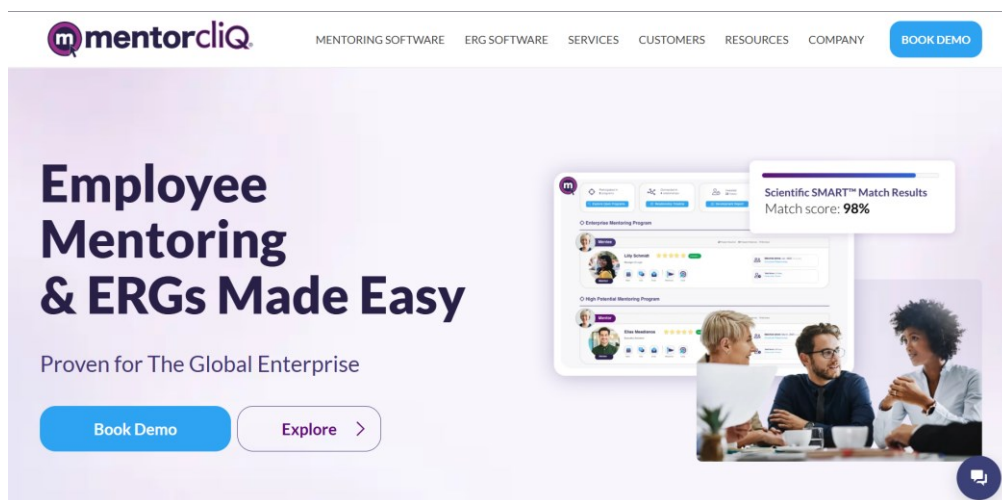


Figure 7 MentorcliQ interface

Table 4 Comparison table between systems

Features	HRDF	MentorCity	Graduway	LinkedIn Learning	Qooper	MentorcliQ	ASPIRA
Type of the system	website	website and mobile app	website and mobile app	website and mobile app	website	website	website
AI chatbot							✓
Course Recommendations				✓	✓	✓	✓
Group session booking		✓	✓		✓	✓	✓
Access to learning resources	✓		✓	✓	✓	✓	✓
Private Session Booking		✓			✓	✓	✓
View Session History		✓	✓		✓	✓	✓
CV builder							✓
Courses center				✓			✓
For free							✓



The assessment of competitive products examines current mentorship and career development platforms to reveal opportunities for ASPIRA's growth. The research team analyzed HRDF together with MentorCity and Alumni Networks and LinkedIn Learning and Qooper and MentorcliQ to identify their functionalities and weaknesses.

The main goal of HRDF involves finding employment opportunities while it does not assist students with educational mentoring. MentorCity allows mentors and mentees to find each other while offering goal-management capabilities beyond career mentoring support. The networking platform Alumni Networks enables university graduates to communicate yet does not provide either career guidance or job placement assistance. LinkedIn Learning offers customized learning material suggestions but fails to provide mentorship support and career development assistance.

Qooper and MentorcliQ implement AI to match mentoring relationships yet their main user base consists of corporate mentoring. ASPIRA stands out through its implementation of artificial intelligence mentor matching services which pairs with individualized professional development resources and academic counseling and specific functions to recommend courses and construct CV builder as well as provide free mentoring materials.

A comparison sheet demonstrates that ASPIRA outperforms its competitors by offering an all-encompassing system for AI-powered mentorship with career development tools and academic support offerings.



chapter 4: **System** **description**



4. System Description

This section outlines the methods used to gather and analyze requirements, including user interviews and questionnaires. These activities were designed to capture both functional and non-functional requirements, enabling us to create a solution that aligns closely with the objectives. It also provides an overview of the system's architecture, key users, and a use case diagram to illustrate interactions between users and the system.

4.1 Users

Mentees: University students or recent graduates seeking career guidance to determine their career paths and acquire the necessary skills, through mentoring.

Educational level: University students or recent graduates with a bachelor's degree.

Experience: Limited or no professional experience, may lack direct practical experience in the job market.

Technical expertise: Basic to intermediate technical skills, beneficiaries are expected to be familiar with using electronic platforms.

Mentors: Mentors are professionals with extensive experience in their fields of work, providing support and guidance to mentees, their primary role is to share their practical experiences to guide mentees towards achieving their career goals.

Educational level: Professionals with advanced degrees or significant experience in their specialized fields.

Experience: Experienced individuals with years of practical knowledge in specific industries or fields.

Technical expertise: Intermediate to advanced technical skills, mentors must be able to use platforms to manage their profiles, interact with mentees, and deliver mentoring sessions.



Admin: are responsible for reviewing mentors registration requests and either approving or rejecting them based on the mentor's submitted CV. Their primary role is to ensure that qualified mentors join the platform to provide high-quality guidance to mentees.

Educational level: Individuals with a solid educational background, typically holding a bachelor's degree or higher, preferably in fields related to management, HR, or administration.

Experience: Experience in administrative or decision-making roles, with the ability to assess qualifications and professional resumes.

Technical expertise: Intermediate to advanced technical skills, including familiarity with electronic platforms, and the ability to navigate the system for reviewing and processing mentor applications.

4.2 Requirements Elicitation

We collected extensive information from various sources. Using both questionnaire and interviews with students (undergraduate, fresh graduate) and mentors, we identified the key requirements and organized them by priority to ensure the implementation of the most essential features first.

A total of 50 responses were received from the questionnaire (Appendix B). The questionnaire has different questions, and the participants comprised 94% undergraduate students, 4% experienced professionals, and 2% fresh graduates(within the last 2 years).

The questionnaire focused on understanding the challenges students and fresh graduates face that led them to need guidance and their preferences for mentoring and other features. When we asked about the biggest challenges in career decision-making, 58% reported a lack of guidance, while 38% struggled to identify jobs matching their skills. Regarding their preferred way of receiving career guidance, 60% choose a mix of all mentoring options, while 34% choose one-to-one personalized mentoring. To accommodate these preferences, we ensured that our platform offers various flexible mentoring options which are group and private mentoring. Additionally, 52% of responders preferred virtual meetings, exceeding the

percentage of those who preferred in-person mentoring, highlighting a strong demand for flexible and remote mentoring options, which is a key pillar of our platform. Participants expressed interest in a recommendation system tailored to their career goals, with 64% strongly supporting the idea. To meet this demand, ASPIRA has integrated a recommendation system that suggests relevant courses based on user interests.

We also asked about other provided features like having an AI-powered chatbot answering their queries, most of the answers were very interested in this feature. and rating mentors feature with responses of 64% strongly agreed. In addition to the history of previous sessions feature, 74% saw it extremely useful as a reference. Based on these insights, ASPIRA has incorporated these features into the platform. These additions enhance the overall mentoring experience by providing structured support, accountability, and continuous learning opportunities.

We also asked about mentoring topics they consider most important, and what would catch their attention in the mentor's profile. Reflecting the responses, ASPIRA will ensure that mentor profiles clearly display key details, making it easier for mentees to find the right mentor.

For the interviews(Appendix A), we engaged with three mentors to gain deeper insights into their needs and preferences. We asked them multiple questions, such as their major and inspiration, what strategies they use with Mentos. One of the key questions explored the potential of offering flexible communication methods, such as video calls, in the mentoring process. All mentors interviewees agreed that this feature would significantly enhance the effectiveness of mentoring and make the process more interactive. Another key question asked about the most challenging aspects of communication with mentees. Ghala Alharbi noted, "The most challenging part is understanding what mentees really need, especially when they text me at an unsuitable time and seem angry". Similarly, Shaden Alshehri stated, "For me, the hardest part is when mentees don't communicate regularly or delay responses". Haifa Alsaif added about mentees not being clear and struggling to explain their challenges. These responses highlight that communication is a common challenge for mentors, with issues ranging from clarity and timeliness.



Another question focused on how technology could improve the mentoring process. The mentors highlighted various benefits of incorporating technology into mentoring. Ghala Alharbi noted that “technology makes mentoring easier by helping me stay organized and share resources quickly. It also simplifies connecting and communicating anytime”. while Shaden Alshehri said Technology is "allowing mentors and mentees to collaborate easily, even if they're in different locations.” Lastly Haifa Alsaif answered “technology simplifies mentoring by assisting with tasks like scheduling meetings and staying in touch. It speeds up communication and keeps everything organized and straightforward.”.

Based on the information gathered, it is evident that the users strongly align with the planned features, particularly the recommendation system and the flexible communication methods. The feedback demonstrates the importance of prioritizing tools that enhance communication and mentoring effectiveness. These findings reaffirm the need for a comprehensive platform that bridges gaps in communication, and provides robust support for professional development.

4.3 User Interactions

4.3.1 Use Case Diagram

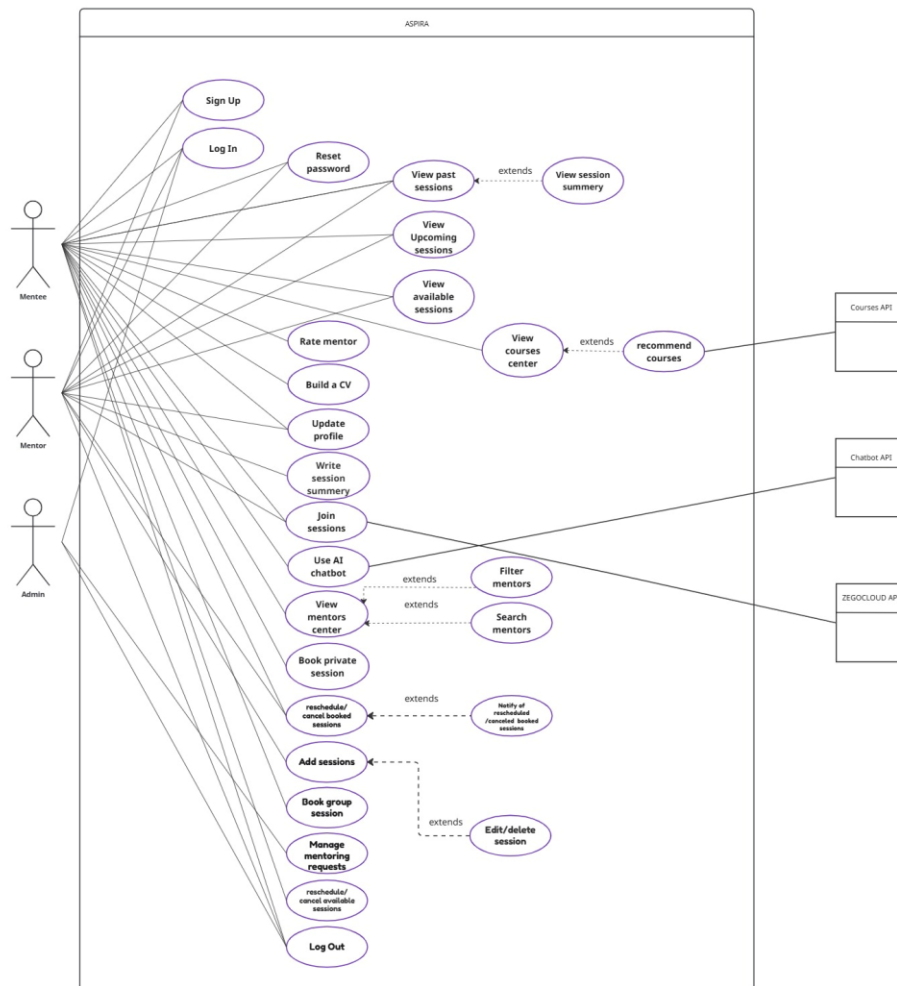


Figure 8 Use Case Diagram



4.4 Product Backlog

Table 5 Product Backlog

ID	PBIs (User Stories)	Size	Type	Status	Acceptance Criteria
1	As an unregistered user (mentee), I want to sign up so that I can create an account and access the website features.	3	Feature	Done	- As an unregistered user(mentee), if I fill in the required fields correctly, then I should successfully create an account. - As an unregistered user (mentee), if I leave any required field empty, then an error message should prompt me to complete the required fields. - As an unregistered user (mentee), if I try to sign up with an email and phone number that is already registered, then an error message should indicate that the email is already in use.
2	As an unregistered user (mentor), I want to sign up so that I can create an account and access the website features.	3	Feature	Done	As an unregistered user(mentor), if I fill in the required fields correctly, then I should wait for the admin to accept me.

					- As an unregistered user (mentor), if I leave any required field empty, then an error message should prompt me to complete the required fields. - As an unregistered user (mentor), if I try to sign up with an email that is already registered, then an error message should indicate that the email is already in use.
3	As a registered user (mentor, mentee, admin), I want to login so that I can use website features.	5	Feature	Done	- As a registered user, if I wanted to use the website features, then I should login using the correct email and password. - As a registered user, if I wanted to use the website features, then if I tried to login with incorrect email and/or password an error message should appear
4	As a user (mentor , mentee), I want to be able to reset my password so that I can regain access to my account if I forget it.	2	Feature	Done	- As a user, if I am on the login page and click "Forgot Password", then the system should send a password reset email. - As a user ,if I follow the reset link and enter a new password, Then the



					system should confirm that my password has been successfully reset and update it in the database.
5	As a logged in user (mentor, mentee, admin), I want to log out of the website so that I maintain better control over my account access.	1	Feature	Done	As a logged-in user, if I click on the logout button, then I will be directed to the home page.
6	As an admin, I want to manage the registration of mentors on the website by accepting or rejecting them, so that I can accept mentors with the necessary skills and experience, and reject those whose experience is not suitable for mentoring.	8	Feature	Done	As an admin, if I log into my account, then I should be able to view a list of all mentor registration requests through the requests page. For each request, I should be able to see the required information, such as the CV, then I can make a decision based on their information by clicking on the accept or reject button for the mentor.
7	As a mentee, I want to have a comprehensive overview of all available mentors, including detailed information, so that I can make an informed decision when selecting a mentor.	5	Feature	Done	As a mentee, if I open the mentors center page then I can view a list of all available mentors, along with detailed information such as their names, photos, and areas of experience.

					As a mentee, if I open the mentors center page and there are no available mentors, then I should see a message informing me that no mentors available.
8	As a mentee, I want to filter mentors by field of expertise and experience years so that I can choose the most suitable mentor for my specific needs and queries.	5	Feature	Done	- As a mentee, if I open the mentors center page and see the filter bar on the left of the screen and choose a field of expertise and/or experience years then only those mentors who specialize in that field should appear. - As a mentee, if I filter mentors by field of expertise and/or experience years and there are no available mentors in that field, then an empty page should appear .
9	As a mentee, I want to search for mentors by name so that I can quickly find the mentor I am looking for.	8	Feature	Done	- As a mentee, if I open the mentors center page, and search for a mentor by specific name, then only a mentor with that name will appear. - As a mentee, if I search for a mentor by name and they are not found in the

					mentors center, then an empty page should appear.
10	As a mentor, I want to add private sessions to my schedule so that I can offer personalized guidance to mentees.	7	Feature	Done	- As a mentor, if I go to the add sessions page and select a private session and input the date, time and click on the add button then the session should be saved and visible in available session. - As a mentor, I should not be allowed to add a new session if the time gap between it and an already existing session is less than one hour.
11	As a mentee, I want to book a private session so that I receive personalized advice in a manner that aligns with my preferences.	8	Feature	Done	- As a mentee, if I open the mentors center page and see a list of all available mentors and I select a suitable mentor then I can see all available time slots for a private session with that mentor. - As a mentee, if I choose the most suitable time for me, and click on the book button then the session will be added in the upcoming sessions, after

					that I should receive a confirmation message for this booking. As a mentee, if no time slots are available with the mentor, then a message will appear stating “No available sessions at the moment” .
12	As a user (mentor or mentee), I want to be able to join a private session so that I can participate in personalized, one-on-one guidance tailored to my needs.	2	Feature	Done	- As a user, if I want to join my private session, I go to the Upcoming Sessions page and click on the Join button. Then will only be able to join the session once its scheduled time has started. - As a user, if I try to join a private session before its scheduled start time, Then I should see a message saying the session hasn't started yet. Please wait!. - As a user, if the scheduled time for my private session has passed and I didn't join it, then I should see a message saying 'Session Ended' to inform me that the session is no longer available.

13	As a mentee, I want to rate mentors after sessions so that I can provide feedback and help improve the ASPIRA platform reliability for others.	3	Feature	Done	- As a mentee, if I complete a session, then I can rate the session with that mentor in past session page. - As a mentee, if I provide a rating (e.g., 1–5 stars), then the system successfully stores it in the database. - As a mentee, if I leave feedback in the text box, then the system saves my comments along with the rating.
14	As a mentee, I want to see a list of my upcoming sessions so that I stay informed, and keep track of my sessions.	5	Feature	Done	- As a mentee, if I open the upcoming sessions page then I can view a list of my currently booked sessions, including details such as the session date,time, mentor's name. - As a mentee, If I have no upcoming booked sessions, then a message should appear stating, "No upcoming sessions".

15	As a mentor, I want to see my upcoming sessions with mentees, so that I stay informed, and keep track of upcoming sessions.	5	Feature	Done	- As a mentor, if I open the upcoming Sessions, then I will be able to see all my upcoming sessions booked by mentees with details such as the session date, time and the name of the mentee who booked it. - As a mentor, If I have no upcoming booked sessions by mentees, then a message should appear stating, “No upcoming sessions”.
16	As a mentor, I want to see a history of my past sessions with mentees, so that I can refer back to them if I need to.	3	Feature	Done	- As a mentor, if I open the past session page then I will be able to see all my past sessions with mentees with details such as the date, time of the session, name of the mentee who booked it. - As a mentor, If I have no past sessions, then a message should appear stating, “No past sessions”.

17	As a mentee, I want to edit the scheduled time of my private session so that I can attend at a time that is convenient for me.	5	Feature	To Do	- As a mentee, if I go to the upcoming sessions, and click on private sessions and select the session I want to reschedule and click on edit button and all available time slots with the mentor will display then I can select a new time and click book button, a confirmation message will appear. - As a mentee, if I didn't find other time slots available with the mentor, then a message would appear stating, "No available sessions at the moment".
18	As a mentee, I want to delete my private or group sessions so that I can avoid attending a session if I no longer need it.	5	Feature	To Do	As a mentee, if I go to upcoming sessions, I can select the session whether (group or private) I want to delete and click on the delete button then a confirmation message for deleting should appear .
19	As a mentee, I want to book available group sessions so that I can gain broader career advice and learn from shared experiences.	5	Feature	To Do	As a mentee, if I open the mentee homepage then I can view a list of all available group sessions with their details, such as



					title, date, time, mentor name and i can choose the group session I want, and I can click on the book button to add it to the upcoming sessions then I should receive a confirmation message for this booking. As a mentee, if no group sessions are available, then I see a message that says, "No group sessions available at the moment."
20	As a mentor, I want to edit an available private session to a time that better fits my availability so that I can still provide quality guidance to the mentee.	5	Feature	To Do	As a mentor, if I go to the available sessions page and select the private session I want to reschedule and click on the edit button then I can update the session's date and time by entering new values in the provided fields and then clicking on the edit button.

21	As a mentor, I want to edit a booked private session to a time that better fits my availability so that I can still provide quality guidance to the mentee.	8	Feature	To Do	As a mentor, if I go to the upcoming sessions page and select the private session I want to reschedule and click on the edit button then I can update the session's date and time by entering new values in the provided fields and then clicking on the edit button, Then a notification will be sent to the mentee whose session times i have modified.
22	As a mentor, I want to add group sessions so that I can share knowledge and expertise with multiple mentees.	3	Feature	To Do	As a mentor, if I go to the add sessions page and select group sessions then I can input title, date, time and click on the add button, then the session should be saved and visible in the available session page ,Then it appears in the home page for mentees.
23	As a mentor, I want to edit a group session to a time that better fits my availability, ensuring I can still effectively deliver the session to mentees.	5	Feature	To Do	As a mentor, if I go to the upcoming sessions page and select the group session I want to reschedule and click on the edit button then I can update the session's title,

					date, and time by entering new values in the provided fields, and then click on the edit button.
24	As a mentor, I want to delete a group session , if I no longer want them, so that I do not have an appointment with mentees if I cannot or do not want to start the session.	5	Feature	To Do	As a mentor, if I go to the upcoming sessions page and select the group session I want to delete and click on the delete button then the group session will be deleted.
25	As a mentor, I want to delete an available private session , if I no longer want them, so that I do not have an appointment with mentees if I cannot or do not want to start the session.	5	Feature	To Do	As a mentor, if I go to the available sessions page then I can select the private session I want to delete and click on the delete button , then a confirmation message for deleting should appear .
26	As a mentor, I want to delete a booked private session, if I no longer want them, so that I do not have an appointment with mentees if I cannot or do not want to start the session.	8	Feature	To Do	As a mentor, if I go to the upcoming sessions ,then I can select the session I want to delete and click on the delete button ,a confirmation message for deleting should appear ,Then a notification will be sent to the mentee whose session times I have deleted.

27	As a user (mentor, mentee), I want to edit my profile information so that I can keep it updated.	3	Feature	To Do	- As a user, if I go to the account page and click on the edit button to change my information then my information should be save and updated - As a user, if I do not make any changes to my information then the original information should remain unchanged
28	As a mentee, I want to use a CV builder so that I can create a professional CV to boost my career growth opportunities.	5	Feature	To Do	- As a mentee, if I go to the CV builder page, then I should see a form that allows me to input my relevant information. - As a mentee, if I complete the form and submit it, then a professional CV should be generated that I can preview and download.
29	As a mentee, I want to be notified when my booked private/group session is rescheduled or deleted by my mentor so that I can stay informed about any changes.	7	Feature	To Do	As a mentee, if my mentor reschedules my booked private/group session, then I should receive a notification so that I can book a replacement session accordingly.

					As a mentee, if my mentor deletes the booked private/group session, then I should receive a notification informing me of the cancellation.
30	As a mentor, I want to write a summary of my past sessions with mentees, so that the mentee can benefit from the key points discussed.	3	Feature	To Do	As a mentor, if I open the past sessions page and click on the summary text box, then I can write key points discussed with the mentee. This summary will also appear on the mentee's Past Sessions page.
31	As a mentor, I would like to be notified whenever a mentee books, modifies, or cancels a private session, so that I will be able to stay informed about my schedule and manage my time effectively.	8	Feature	To Do	- As a mentor , if the mentee reschedules the private session that he booked with me , then I should receive a notification. - As a mentor , if the mentee booked a new private session with me , then I should receive a notification. - As a mentor , if the mentee deletes the private session that he booked with me , then I should receive a notification.

32	As a mentee, I want to ask an AI chatbot about career advising so that I can get the information easily and correctly.	8	Feature	To Do	- As a mentee, if I asked the chatbot a career question then it should answer me. - As a mentee, if I didn't write anything then it should not respond.
33	As a mentee, I want to explore a dedicated courses center that provides information and links to learning resources so that I can easily access relevant educational opportunities.	5	Feature	To Do	- As a mentee, if I open the courses center page, then I can browse a list of available courses. - As a mentee, if I click on a course link, then the course page opens in a new tab.
34	As a mentee, I want to benefit from a course recommendation system that suggests learning materials so that I can find courses that match my interests and preferences.	8	Feature	To Do	As a mentee, if I open the courses center, then I can filter personalized course recommendations based on my interests and preferences.
35	As a mentee, I want the chatbot to answer me within 10 seconds so that I don't have to wait for a long time.	3	Feature	To Do	As a mentee, if I ask a chatbot, then it should answer within 10 seconds.
36	As a mentee, I want to see a history of my past sessions, so that I can refer back to them if I need to.	3	Feature	In progress	As a mentee, if I open past sessions then I can see a list of all my past booked sessions with

					details like the session date, time, mentor name, rate, session summary As a mentee, if I have no past sessions, then I should see a message that says, "No past sessions found."
37	As a user(mentor, mentee,admin), I want the platform to be easy to use and navigate, with intuitive interfaces and clear instructions, so that I can learn how to use it in about 10 minutes.	3	Feature	In progress	As a user, if I open the platform for the first time,then I should learn how to use it in about 10 minutes
38	As a user (mentor, mentee,admin), I want the platform to be available 99% of the time I try to access it, so that I don't get frustrated.	3	Feature	In progress	As a user, if I want to use the site, then it must be available 99% of the time I try to access it.
39	As a mentee, I want the results of my search for a mentor to appear within 10 seconds, so that I can ensure a fast and responsive experience.	3	Feature	In progress	As a mentee, if I type the mentor's name in the search box, then the results should appear within 10 seconds.



40	As a user(mentor, mentee), I want the session history to appear within 10 seconds, so that I can see my upcoming and past sessions quickly and efficiently.	3	Feature	In progress	As a user, if I go to the upcoming or past session history page, then the data should appear within 10 seconds.
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Table 5 Shows the status of user stories for each release.

All user stories related to Release 1 have been successfully implemented, and all acceptance criteria have been met , their status is marked as "Done".

As for the stories planned for Release 2, their status is currently "To Do".

The status "In Progress" indicates that the story was implemented in this release, but work on it will continue in the release2.



chapter 5: **System design**

5. System Design

5.1 Architecture

The Client-Server architecture has been selected as the most suitable design for the ASPIRA platform, as it provides centralized data management and ensures effective communication between users (mentees, mentors, and admin) and the server. The server is responsible for processing all client requests, including searching for mentors by name, filtering by field, scheduling sessions, recommending relevant courses, and reviewing mentor registration requests submitted by mentors for approval or rejection by the admin. It also connects to the database to store and retrieve information before sending the results back to the clients using the appropriate protocol.

This architecture significantly simplifies data management through centralized administration, allowing the storage of all critical information such as mentor registration statuses in a single database, making it easier for the admin to review and manage the system efficiently. Additionally, all records remain easily accessible since they are maintained on the central server, allowing for immediate data retrieval when needed.

This architecture also ensures fast response times when searching for mentors or filtering results, as the server quickly processes and returns data without requiring local storage on user devices, improving platform performance. Moreover, it provides strong security and data protection, ensuring that sensitive information such as registration requests, CVs, and session details is stored in a secure location rather than being distributed across multiple devices, reducing the risk of data loss or unauthorized access.

Furthermore, the Client-Server architecture offers scalability, allowing the server capacity to be expanded easily to handle increased requests as the number of users grows, ensuring system stability and reliability even with higher traffic.

Based on these factors, the Client-Server architecture is the ideal choice for ASPIRA in terms of performance, reliability, and ease of use, enabling users (mentees, mentors, and admin) to interact with the platform efficiently while ensuring a smooth and dependable experience for all.

The following diagram illustrates the Client-Server architecture chosen for designing the ASPIRA platform.

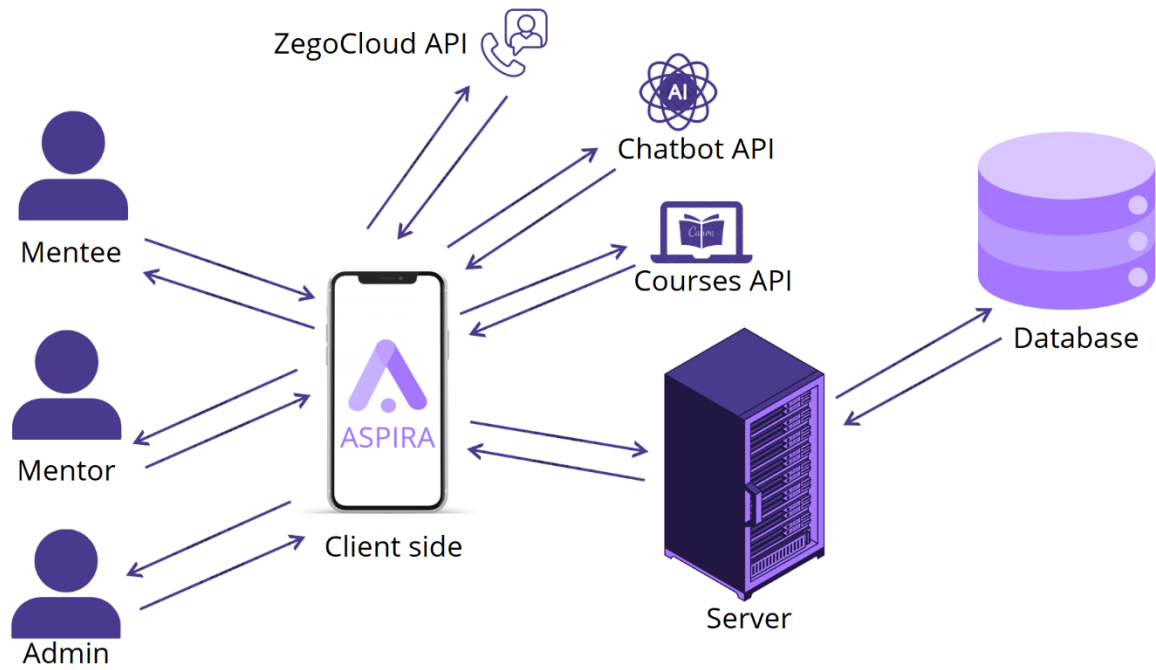


Figure 9 Client-Server architecture



5.2 Class Diagram /DFD

Figure 10 shows the class diagram of the ASPIRA platform. There are three main user types: Admins, Mentors, and Mentees, all extending from the general User class. Mentors can sign up, add and join sessions, view notifications, update their profile, upload certificates, and write summaries after each session. Mentees can sign up, search and filter mentors, book and join sessions, view notifications, build their CVs through the CVbuilder, and access recommended courses via the courseCenter. Admins handle mentoring requests in the background to ensure the platform runs smoothly.

ASPIRA also includes core features like Sessions, Notifications, Rating, CVbuilder, and CourseCenter. Notifications keep users updated, the rating allows mentees to leave feedback, and sessions are managed through a structured and accessible flow.

All these components work together to make mentorship easier, smarter, and more effective.

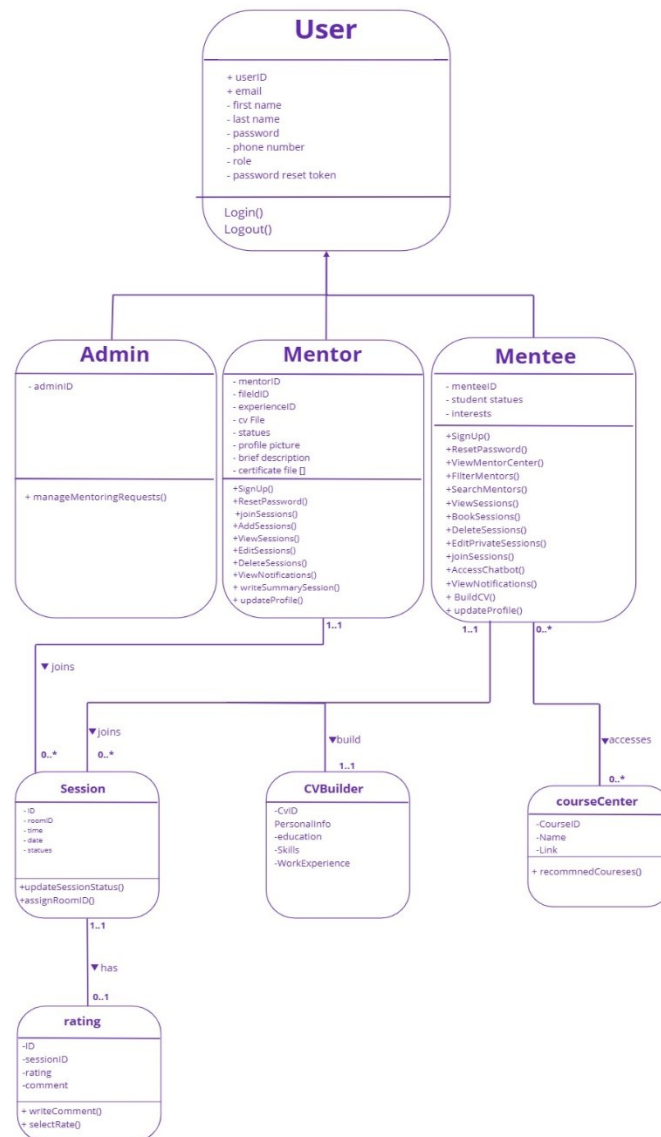


Figure 10 The ASPIRA Platform Class Diagram

5.3 Component Level Design

To better understand how ASPIRA works, we've illustrated three of its most important features: **Rate Mentors**, **Add Private Session**, and **Search Mentors**. We've included flowcharts for both the Search and Add Private Session features to show how each step flows from the moment a mentor or mentee takes action, to how the system responds. For the Rate Mentors feature, which involves more conditional logic (like checking if a session was completed or if a mentor was already rated), we used pseudocode to clearly represent the process. These visual and logical breakdowns offer a simple, structured look at how these



features function and help deliver a smooth, personalized experience for both mentors and mentees on the platform.

5.3.1 Rate mentors Pseudocode

Classification: Function

Definition: The mentee can rate mentors after a session is completed. Each session can be rated once by the mentee using a 5-star scale and comment. Ratings will be saved and used to update the mentor's average rating, which is displayed in the Mentor Center.

Pre-condition: The user must be a mentee, logged in, and viewing the Past Sessions page with completed sessions.

Post-condition: The rating and comment are saved, mentor's average rating is updated, and the rating inputs become disabled (read-only).

Pseudocode:

1. START
2. currentUser = Get logged-in user.
3. pastSessions = Retrieve all completed sessions where mentee_id == currentUser.id
4. FOR each session in pastSessions DO
5. IF session.is_rated == TRUE THEN
6. Display session details with existing rating and comment as read-only
7. ELSE
8. Display session details with:
 - Comment input field
 - Star rating input (1 to 5)
9. IF user clicks Submit THEN
10. selectedRating = Get selected star rating



11. commentText = Get comment input
12. Save (mentee_id, mentor_id, session_id, selectedRating, commentText) to SessionRatings table
13. Mark session as rated (session.is_rated = TRUE)
14. mentorRatings = Retrieve all ratings where mentor_id == session.mentor_id
15. avgRating = Calculate average of mentorRatings
16. Update MentorProfile table: set avg_rating = avgRating
17. Disable rating and comment inputs for this session (make read-only)
18. END IF
19. END IF
20. END FOR
21. Display all past sessions with ratings/comments accordingly
22. END

5.3.2 Add Private session Flowchart

Figure 11 illustrates the flowchart of the **Add Private Session** functionality for mentors in the ASPIRA platform. Mentors begin by accessing the "Add Sessions" tab, where they can input the date and time of a new session they wish to offer. The system checks that each session has at least a one-hour gap from the previous session by the same mentor. If there is no time overlapping, then the session will be saved to the database and added to the list of available sessions. From there, the system continues to monitor whether the session gets booked by a mentee. If booked, it is automatically moved to the mentor's upcoming sessions list; otherwise, it remains available for mentees to book. This process ensures that mentors have full control over their scheduling while allowing mentees to easily discover and reserve suitable session slots.

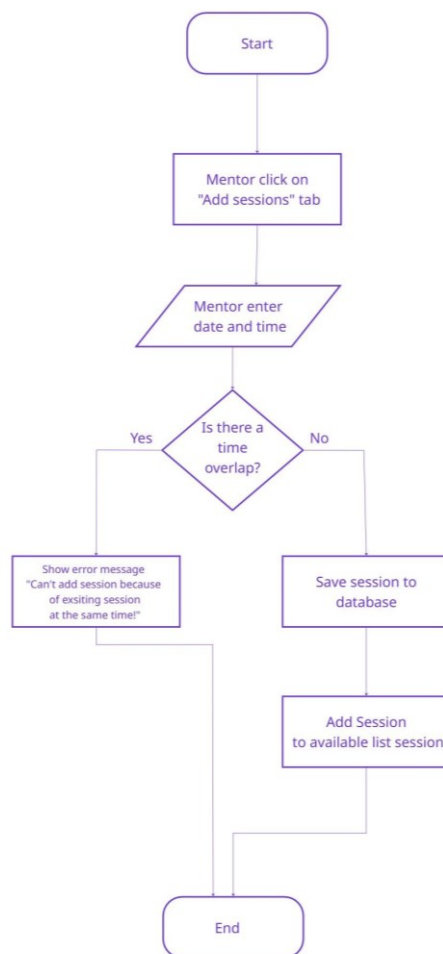


Figure 11 Add Private session Flowchart

5.3.3 Search mentors Flowchart

Figure 12 shows how the **Search Mentors** feature works in the ASPIRA platform. When a mentee opens the Mentor Center, they can type in the name of the mentor they're looking for. The system will then check if a mentor with that name exists. If it finds a match, the mentor's profile is shown. This feature is designed to make it quick and easy for mentees to find specific mentors, helping them connect with the right guidance more efficiently.

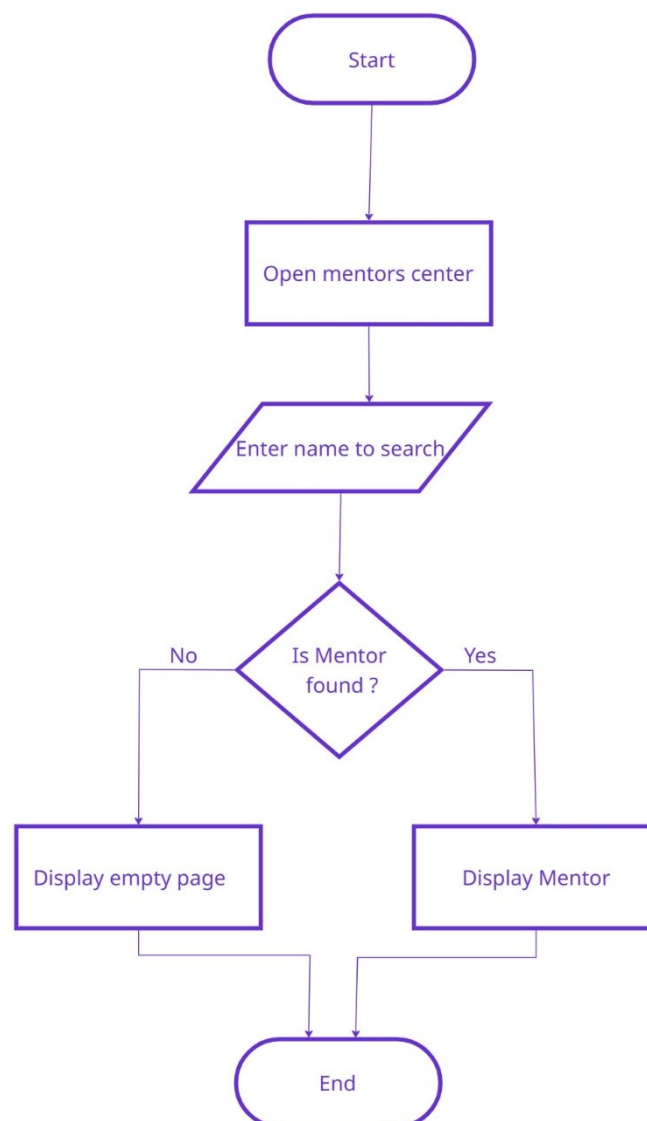


Figure 12 Saerch mentors Flowchart

5.4 Data Design

This section provides an overview of the data design for the first release of the project, beginning with an Enhanced Entity-Relationship diagram (EER). The EER diagram illustrates the structure of the entities and their relationships within the system. Following this, the section presents the Relational Schema, which outlines how data is organized into tables, specifying the relationships and constraints between them. The section also includes the Data Dictionary, an essential component that describes each entity in the database, including its name, attributes, and any relationships it has with other entities. Finally, it concludes with a section on EER considerations, which highlights the specific aspects of the ASPIRA platform that were taken into account when designing this EER.

5.4.1 Data Models

5.4.1.1 EER Diagram

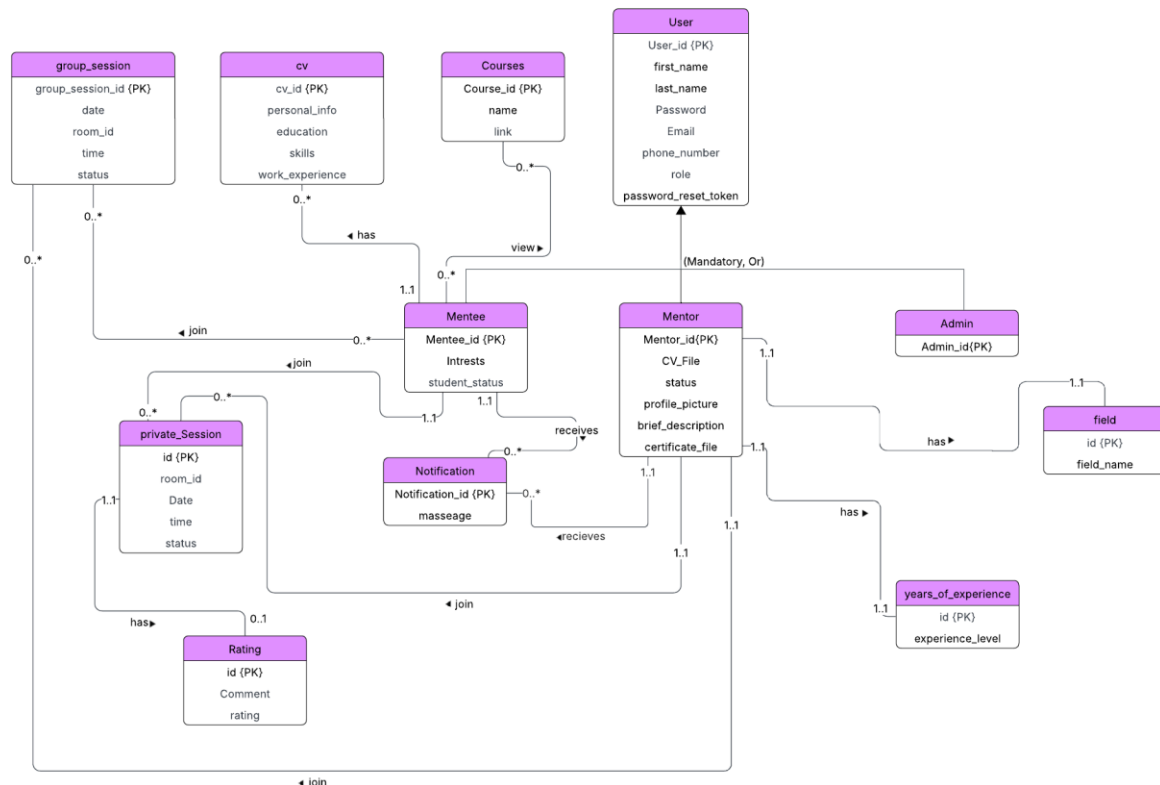


Figure 13 EER



5.4.1.2 Relational Schema

User {User_id, first_name, last_name, password, email, phone_number, role, password_reset_token}

Primary Key: User_id

Admin {Admin_id, user_id}

Primary Key: Admin_id

Foreign Key: user_id

Mentor {Mentor_id, cv_file, status, profile_picture, brief_description, certificate_file, user_id, field_id, experience_id}

Primary Key: Mentor_id

Foreign Key: user_id, field_id, experience_id

Mentee {Mentee_id, interests, student_status, user_id}

Primary Key: Mentee_id

Foreign Key: user_id

Field {id, field_name}

Primary Key: id

Years_of_Experience {id, experience_level}

Primary Key: id



private_Session {id, room_id, date, time, status, Mentor_id, Mentee_id}

Primary Key: id

Foreign Key: Mentor_id, Mentee_id

Group_Session {group_session_id, date, room_id, time, status, Mentor_id, Mentee_id}

Primary Key: group_session_id

Foreign Key: Mentor_id, Mentee_id

Rating {id, comment, rating, Session_id, Mentor_id, Mentee_id}

Primary Key: id

Foreign Key: Session_id

Notification {Notification_id, message, Mentee_id, Mentor_id}

Primary Key: Notification_id

Foreign Key: Mentee_id, Mentor_id

cv {cv_id, personal_info, education, skills, work_experience, Mentee_id}

Primary Key: cv_id

Foreign Key: Mentee_id

Courses {course_id, name, link, Mentee_id}

Primary Key: course_id

Foreign Key: Mentee_id

5.4.1.3 Data Dictionary

5.4.1.3.1 Data Dictionary of Entities

Table 6 Data Dictionary of Entities

Entity Name	Description	Occurrence
admins	A user who reviews mentor registration requests and approves or rejects them based on qualifications.	The admin can approve or reject multiple mentor
field	Represents areas of expertise (e.g., programming, design, marketing) for mentors.	Each field appears once and may be selected by many mentors.
mentees	A user who receives mentorship, Can join sessions, view courses, use the CV builder and use AI chatbot.	A mentee is allowed to join multiple private or group sessions
mentors	A user who provides mentorship sessions.	A mentor is allowed to guide many mentees.
ratings	Allows mentees to rate sessions with a scale (1-5) and a comment.	One rating is submitted by the mentee for each completed session with a mentor.
private_sessions	A private session that takes place between one mentor and one mentee	Each mentor or mentee can participate in one, none, or many sessions, but not simultaneously in more than one session

users	The actual user of the platform.	A user Can be a mentor, mentee, or admin.
years_of_experience	Provides years of experience for mentors.	Each experience can be selected by multiple mentors.
notifications	Sends notification when sessions are modified or canceled by either mentor or mentee.	Each user (mentor or mentee) can receive one, none or many notifications.
courses	Educational resources recommended to mentees. Contains a name and a link.	Each course can be viewed by one , none or many mentees.
group_sessions	Represents group mentoring sessions initiated by a mentor and attended by multiple mentees	Each group session is initiated by one mentor and attended by one, none or many mentees.
cv	A CV builder used by the mentee to create a CV by entering personal information, education, skills, and work experience..	Each mentee can create one, none or many CVs.

5.4.1.3.2 Data Dictionary of Relationships

Table 7 Data Dictionary of Relationships

Entity Name	Multiplicity	Relationship	Entity Name	Multiplicity
User	1..1	has	Mentor	0..*
User	1..1	has	Mentee	0..*
User	1..1	has	Admin	1..1
Mentor	1..1	has	field	1..1
Mentor	1..1	has	years_of_experience	1..1
Mentor	1..1	receives	Notification	0..*
Mentee	1..1	receives	Notification	0..*
Mentee	0..*	view	Courses	0..*
group_Session	0..*	join	Mentee	0..*
Group_Session	0..*	join	Mentor	1..1
private_Session	0..*	join	Mentee	1..1
private_Session	0..*	join	Mentor	1..1
private_Session	1..1	has	Rating	0..1
cv	0..*	has	Mentee	1..1



5.4.1.3.3 Data Dictionary of Attributes

Table 8 Data Dictionary of Attributes

Entity Name	Attribute	Description	Data Type	Length	Nulls	Mu lti- val ued	Default Value	Range	PK
admins	admin_id	Unique identifier for the admin	int	11	N	N			PK
field	id	Unique identifier	int	11	N	N			PK
	field_name	Name of the field or specialization	varchar	255	N	N	CHARSET= utf8;		
mentees	mentee_id	Unique identifier for the mentee	int	10	N	N			PK
	student_status	Current academic level of the mentee	enum		N	N			
	interests	Textual input describing the mentee's interests	text		N	N	CHARSET= utf8; -- --		



mentors	mentor_id	Unique identifier for the mentor	int	11	N	N			PK
	cv_file	Path to mentor's uploaded CV	varchar	255	N	N			
	status	Current approval status	enum		Y	N	'pending'		
	profile_picture	Path to mentor's profile picture	varchar	255	Y	N	NULL		
	brief_description	Short bio of the mentor	text		Y	N			
	certificate_file	Path to uploaded certificates	longtext		Y	N	CHARSET=utf8;		
ratings	id	Unique identifier	int	10	N	N			PK
	rating	Numerical score given to session	int	11	Y	N	NULL		



	comment	Feedback comment on session	text		Y	N	CHARSET= utf8mb4		
private_sessions	id	Unique identifier	int	11	N	N			PK
	date	Scheduled date of session	date		N	N			
	time	Scheduled time of session	time		N	N			
	room_id	Generated room identifier	varchar	255	Y	N	NULL		
	status	Current approval status	enum		Y	N	'available'		
users	user_id	Reference to the user	int	11	N	N			PK
	first_name	User's first name	varchar	50	N	N			
	last_name	User's last name	varchar	50	N	N			



	email	Email address	varchar	100	N	N			
	password	Encrypted password	varchar	255	N	N			
	phone_number	User's contact number	varchar	20	N	N			
	role	User role	enum		N	N			
	password_reset_token	Token for resetting password	varchar	255	Y	N	NULL		
years_of_experience	id	Unique identifier	int	11	N	N			PK
	experience_level	Years of experience level	varchar	50	N	N	CHARSET=utf8;		
group_session	group_session_id	Unique ID for group session	int	11	N	N			PK
	date	Date of the session	date		N	N			
	room_id	Generated room identifier	varchar	255	Y	N			



	time	Time of the session	time		N	N			
	status	Booking status	enum('available', 'booked', 'completed')		Y	N	'available'		
cv	cv_id	Unique identifier for the CV	int	11	N	N			PK
	personal_info	Basic personal details	text		Y	N			
	education	Education history	text		Y	N			
	skills	List of skills	text		Y	N			
	work_experience	Previous job experience	text		Y	N			
Courses	course_id	Unique identifier for each course	int	11	N	N			PK
	name	Name of the course	varchar	100	N	N			

	link	Online URL for the course	varchar	255	Y	N			
Notification	notification_id	Unique identifier for notification	int	11	N	N			PK
	message	Content of the notification	text		Y	N			

5.5 Interface Design

5.5.1 Sitemap

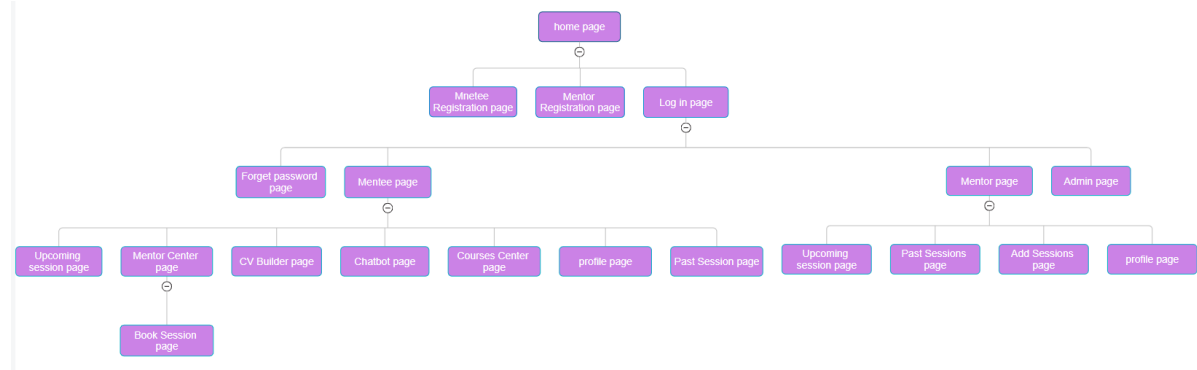


Figure 14 sitemap

5.5.2 UX guidelines

In developing the ASPIRA platform, we prioritized five core User Experience (UX) guidelines to ensure that our platform is intuitive, engaging, and accessible. These principles of error prevention, design dialog to yield closure, informative feedback, striving for consistency, and reducing short-term memory load were carefully applied to enhance the

overall user journey. Each plays a critical role in making ASPIRA a platform that is not only functional but also user-friendly and reliable.

1. Error Prevention

To minimize user errors and improve data accuracy, ASPIRA uses clear guidance, visual cues, and validation checks.

- Inputs like email, and numeric fields are validated in real time to prevent incorrect submissions.
- When submitting a form, if a required field is left empty, the input box is highlighted in red to signal missing information.
- If a user enters invalid content, like typing numbers in a name field, a specific error message appears below the field, guiding them to correct it, as shown in figure [15].

These measures help ensure both data integrity and user confidence, reducing frustration and improving the platform's usability.

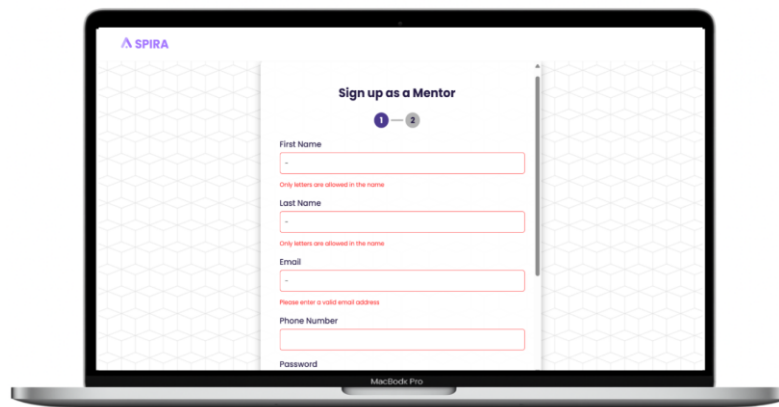


Figure 15 Error Prevention

2. Design Dialog to Yield Closure

ASPIRA ensures users receive confirmation when completing important tasks, promoting a sense of accomplishment and clarity.

- When a mentee successfully books a popup message is shown confirming the action, as shown in figure [16].
- Before accepting or rejecting mentor requests, admins are shown a confirmation prompt to verify their action.

These dialogs guide users through the platform, eliminating uncertainty and reinforcing completed actions.

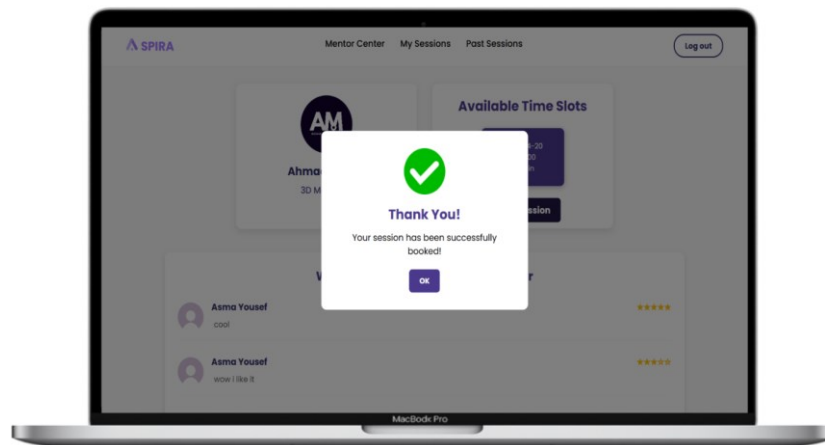


Figure 16 Design Dialog to Yield Closure

3. Informative Feedback

To support smooth interaction, ASPIRA provides real-time feedback across all major touchpoints.

- During login, users are notified instantly if credentials are incorrect or if their account is pending admin approval.
- During private session scheduling, if a mentor attempts to add a session less than one hour before an already scheduled session, a clear error message appears to prevent booking conflicts, as shown in figure [17].

This continuous feedback ensures users always understand what's happening and what to do next.

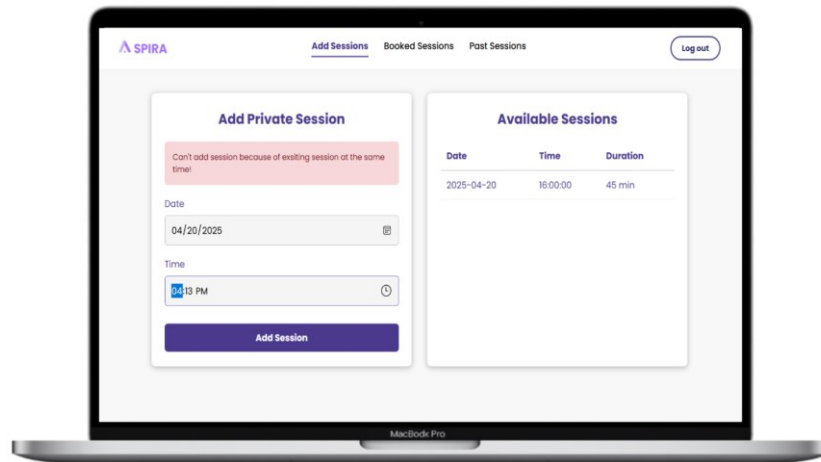


Figure 17 Informative Feedback

4. Striving for Consistency

ASPIRA maintains a consistent design language across all pages to ensure users can navigate the platform with ease and confidence.

The ASPIRA logo and top navigation bar are fixed and visible on every page, offering seamless access to key sections like Mentor Center. This consistent placement reinforces orientation and usability.

All interactive elements such as the “Book,” “Approve,” “Reject,” buttons follow a standardized design:

- Primary actions are colored in purple or green.
- Destructive actions like “Reject” are always in red, as shown in figure [18].
- Buttons respond with clear hover effects and confirmation dialogs to maintain uniform feedback.

ASPIRA keeps a consistent look and feel across the platform to make navigation easier. Mentor cards and course cards follow similar clean layouts, with key info shown up front. While sign-up forms for mentors and mentees have different fields, they share the same structure and validation style helping users feel familiar with the flow no matter where they are.

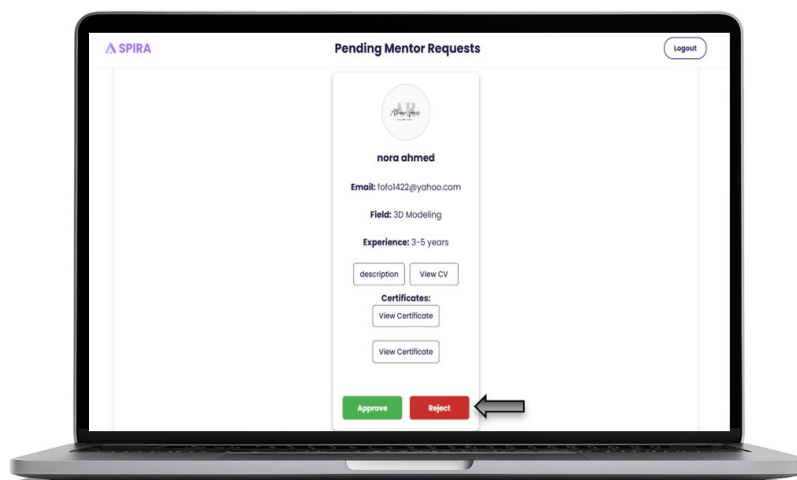


Figure 18 Striving for Consistency

5. Reducing Short-Term Memory Load

Human short-term memory is limited, and effective UX design must minimize the cognitive effort required to remember details. ASPIRA addresses this by ensuring that essential information remains visible, accessible, and well-organized throughout the platform.

Implemented Features in ASPIRA:

- Upcoming Sessions Overview:

On the mentee dashboard, all booked sessions are displayed along with date and time. This reduces the need for users to memorize upcoming appointments and ensures they're always aware of their schedules, as shown in figure [19].

- Mentor Center Quick View:

Mentor profiles show essential information upfront field of expertise, years of experience, and availability so mentees can make decisions quickly without needing to open multiple profiles or retain complex comparisons in memory.

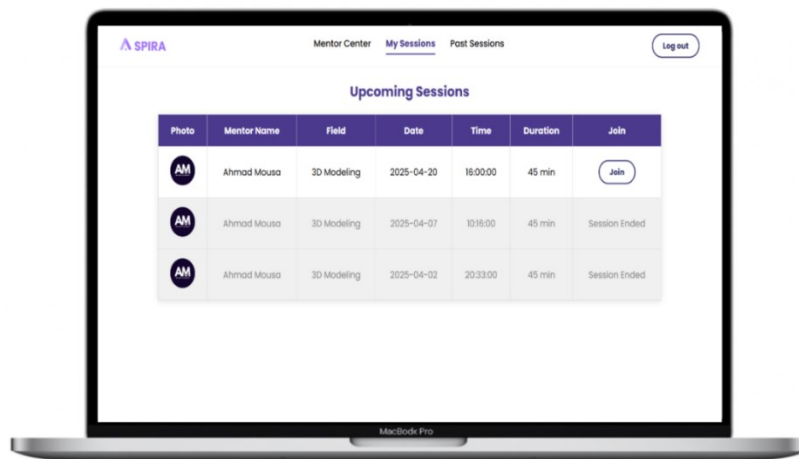


Figure 19 Reducing Short-Term Memory Load



chapter 6: System implementation



6. System Implementation

The system implementation phase of ASPIRA involved the development of four key components: a web application for mentees, a mentor interface to manage their sessions, an admin page for accepting or rejecting mentor applications, and a backend system with a connected database to support all three. In this section, we discuss the key implementation steps and procedures, as well as the major and challenging parts of the code.

6.1 Implementation Key Steps

The implementation of our ASPIRA platform involved several key steps, including:

- Requirements Gathering

We needed to gather the requirements to lay the foundation of the development process. This included understanding the project objectives, user needs, and technical requirements.

- Designing the User Interface

The user interface was designed with a focus on intuitive navigation and a clean, organized layout. Key areas were carefully planned to ensure a smooth and user-friendly experience for all users.

- Development

The development phase included building the front-end, back-end, and the supporting database structure:



1. Building the Front-End :

We built the front-end of the platform using HTML, CSS, and JavaScript. This included designing and developing forms and interactive components to ensure a responsive, user-friendly, and visually appealing interface.

2. Building the Back-End:

We started by creating the database. We used phpMyAdmin, a popular web-based database management tool, to manage our SQL database. This allowed us to store and retrieve data efficiently. Then, we developed the back-end using PHP for the website. This involved creating server-side components such as Database Management, data validation, authentication, and authorization logic.

- Testing and Debugging

Throughout the development process, we performed thorough testing and debugging on both the front-end and back-end components. This included testing individual features based on user stories, conducting integration tests to ensure seamless interaction between system modules, and performing user acceptance testing after implementation to confirm that the platform meets user expectations and functions correctly.

6.2 Major Parts of ASPIRA Code

This section outlines the major implementation steps behind the core functionalities of the ASPIRA platform. These components represent the backbone of the system and are essential to delivering a smooth and responsive mentorship experience. Key parts include the private session creation and booking features, real-time video call integration powered by the ZEGOCLOUD API, and secure email functionality implemented through PHPMailer.

6.2.1 Add Session Function

The "Add Private Session" feature allows mentors to create and manage their availability by adding private session slots directly from the interface. This functionality is implemented using PHP and MySQL, with form validation and scheduling logic to ensure proper session management.

Figure 20 shows the flow that begins when the page loads. The system checks whether the mentor is logged in by verifying the presence of a valid `user\_id` in the active session. If the user is not authenticated, they are redirected to the login page. Once authenticated, mentors can fill out a form to add a session by selecting a date and time. The system ensures that all required fields are filled before processing the request.

```
8  if (!isset($_SESSION['user_id'])) {  
9      header("Location: login.php");  
10     exit();  
11 }
```

Figure 20 check login

Upon submission, the server checks whether there is any time conflict with existing sessions on the same date. This is handled by querying the database for all sessions associated with the same mentor and date. By retrieving only the relevant session times, the system ensures that validation is efficient and accurate. Figure 21 illustrates how both the newly requested session time and the existing ones are converted to UNIX timestamps using the `strtotime()` function. The time difference is then calculated in hours using the expression `abs($new_time - $existing_time) / 3600`, which allows the system to evaluate whether the requested session overlaps with or is too close to another.

To maintain a buffer between sessions, a condition is used to check if the time difference is less than one hour (if `$time_difference < 1`). If this condition is true, the system flags the entry as a conflict and prevents the session from being added. This logic is essential for avoiding double bookings and ensuring that mentors have enough preparation and recovery time between sessions. If a conflict is detected, a clear error message is displayed to the mentor, allowing them to select a different time slot. This conflict detection mechanism helps maintain schedule integrity, avoids overbooking, and improves the reliability of session planning on the platform.

```

23
24 // check the conflict
25 $sql_check = "SELECT time FROM sessions WHERE mentor_id = ? AND date = ?";
26 $stmt_check = $conn->prepare($sql_check);
27 $stmt_check->bind_param("is", $mentor_id, $date);
28 $stmt_check->execute();
29 $result_check = $stmt_check->get_result();
30
31 $conflict = false;
32 while ($row = $result_check->fetch_assoc()) {
33     $existing_time = $row['time'];
34
35     // حساب الفرق بين الوقت المدخل والجلسات الموجودة
36     $new_time = strtotime($time);
37     $existing_time = strtotime($existing_time);
38     $time_difference = abs($new_time - $existing_time) / 3600; // تحويل الفرق إلى ساعات
39
40     if ($time_difference < 1) { // أقل من ساعة
41         $conflict = true;
42         break;
43     }
44 }
45 $stmt_check->close();
46
47 if ($conflict) {
48     $message = "Can't add session because of exsiting session at the same time! ";
49 }

```

Figure 21 check conflict the time

Figure 22 shows if no conflict exists, the session is inserted into the sessions table using an INSERT statement, which includes the mentor's ID, selected date, selected time, and sets the session status to 'available'. This ensures that the session becomes immediately visible for booking by mentees. The insertion is handled securely using a prepared statement, where the values are safely bound and executed to prevent SQL injection. The data for this process is collected from an HTML <form> that uses the POST method and includes a hidden input to carry the mentor's ID from the active session. The form also contains an <input type="date"> element for selecting the session date, with the min attribute set to <?php echo date('Y-m-d'); ?>, which restricts users from choosing past dates. Additionally, an <input type="time"> is used to allow mentors to specify the session time in a consistent 24-hour format. These elements ensure that the user submits clean, valid input before the session is processed by the backend.

```

47 if ($conflict) {
48     $message = "Can't add session because of exsiting session at the same time! ";
49 } else {
50     // إدخال الجلسة في قاعدة البيانات
51     $sql_insert = "INSERT INTO sessions (mentor_id, date, time, status) VALUES (?, ?, ?, 'available')";
52     $stmt_insert = $conn->prepare($sql_insert);
53     $stmt_insert->bind_param("iss", $mentor_id, $date, $time);
54
55     if ($stmt_insert->execute()) {
56         $success = true;
57         $message = "Session added successfully!";
58         header("Location: addPrivateSession.php"); // إعادة تحميل الصفحة
59         exit();
60     } else {
61         $message = "Error adding session " . $conn->error;
62     }
63
64     $stmt_insert->close();
65 }

```

Figure 22 php insert the session

```

450 <div class="add-session-form">
451 <h2 class="form-title">Add Private Session</h2>
452
453 <?php if (!empty($message)): ?>
454 <div class="alert"><?php echo $success ? 'alert-success' : 'alert-danger'; ?>>
455 <?php echo $message; ?>
456 </div>
457 <?php endif; ?>
458
459 <form method="POST" action="">
460 <input type="hidden" name="mentor_id" value="<?php echo $_SESSION['user_id']; ?>">
461
462 <div class="form-group">
463 <label for="date">Date</label>
464 <input type="date" id="date" name="date" min="<?php echo date('Y-m-d'); ?>" required>
465 </div>
466
467 <div class="form-group">
468 <label for="time">Time</label>
469 <input type="time" id="time" name="time" required>
470 </div>
471
472 <button type="submit" class="add-btn">Add Session</button>
473 </form>
474 </div>
475
476
477

```

Figure 23 html insert the session

The newly added session appears under the "Available Sessions" section, where only future sessions that are available for booking are displayed. The sessions are organized in chronological order and shown in a clear table format that includes the date, time, and a fixed duration of 45 minutes, which is considered an ideal length for maintaining focus and productivity during mentorship.[23]

6.2.2 Book Session Function

The "Book Private Session" feature allows mentees to view a mentor's profile and select from a list of available time slots to reserve a session. This feature ensures a smooth booking experience through a combination of PHP, SQL, and JavaScript components. As shown in Figure 24 the system executes a SQL query that joins the mentors, users, and field tables to gather the mentor's profile information. This includes their full name, profile picture, field of expertise, and a brief description. These details are then displayed in a neatly formatted profile card, helping the mentee get to know the mentor before booking.

```

26 $sql = "SELECT
27     users.first_name,
28     users.last_name,
29     COALESCE(mentors.profile_picture, '') AS profile_picture,
30     COALESCE(field.field_name, 'Unknown') AS field,
31     COALESCE(mentors.brief_description, 'No description') AS brief_description
32 FROM mentors
33 JOIN users ON mentors.user_id = users.user_id
34 LEFT JOIN field ON mentors.field_id = field.id
35 WHERE mentors.mentor_id = ?";
36
37 $stmt = $conn->prepare($sql);
38 $stmt->bind_param("i", $mentor_id);
39 $stmt->execute();
40 $result = $stmt->get_result();
41 $mentor = $result->fetch_assoc();
42 $stmt->close();

```

Figure 24 SQL query that joins the mentors, users, and field

To display the available time slots, the backend first retrieves the mentor's actual user_id by querying the mentors table using their mentor_id. It then filters sessions from the sessions table where the status is set to 'available' and the combined date and time is in the future, ensuring only upcoming and unbooked slots are shown. These results are ordered chronologically and passed to the frontend.

```
// جلب بيانات من الجلسات
$sql = "SELECT user_id FROM mentors WHERE mentor_id = ?";
$stmt = $conn->prepare($sql);
$stmt->bind_param("i", $mentor_id);
$stmt->execute();
$result = $stmt->get_result();
$mentor_data = $result->fetch_assoc();
$actual_mentor_id = $mentor_data['user_id']; // معرف المستخدم الذي يجب البحث به في الجلسات
$stmt->close();

// جلب user_id على الجلسات بناءً
$sql = "SELECT id, date, time
FROM sessions
WHERE mentor_id = ?
AND status = 'available'
AND CONCAT(date, ' ', time) >= NOW()
ORDER BY date, time";

$stmt = $conn->prepare($sql);
$stmt->bind_param("i", $actual_mentor_id);
$stmt->execute();
$result = $stmt->get_result();
$available_sessions = $result->fetch_all(MYSQLI_ASSOC);
$stmt->close();
```

Figure 25 Retrieve reserved sessions

In Figure 26 each session is displayed as a clickable time slot card containing the session's date, time, and a fixed duration of 45 minutes. The session ID is embedded in a data-id attribute, allowing easy reference in JavaScript. Once the page is fully loaded, an event listener (DOMContentLoaded) is triggered, and each time slot is initialized with a click event. When a mentee clicks on a slot, JavaScript captures the selected session ID using getAttribute('data-id'), visually highlights the slot by toggling a selected class, and makes the "Book Session" button visible by modifying its display property. This interactivity is handled through a loop applied on all elements with the .time-slot class using querySelectorAll(), ensuring a smooth and intuitive selection process.

```
<script>
document.addEventListener('DOMContentLoaded', function() {
  document.querySelectorAll('.time-slot').forEach(slot => {
    slot.addEventListener('click', function() {
      document.querySelectorAll('.time-slot').forEach(s => s.classList.remove('selected'));
      this.classList.add('selected');
      window.selectedSlotId = this.getAttribute('data-id');
      document.getElementById('book-button').style.display = 'block';
    });
  });
});
```

Figure 26 Javascript Retrieve reserved sessions

When the "Book Session" button is clicked, Figure 27 illustrates how JavaScript sends a `fetch()` request to `book_session.php`, using the POST method to pass the selected session ID to the backend. This dynamic request updates the corresponding session's status from 'available' to 'booked' and links it to the current mentee's ID in the database. If the operation is successful, the mentee is shown a confirmation modal with a personalized success message. Figure 28 shows how this is handled by calling the `showModal()` function, which updates the modal's content and displays it using `document.getElementById("successModal").style.display = "flex"`. When the user clicks the "OK" button inside the modal, the `closeModal()` function is triggered, redirecting them to their "Upcoming Sessions" page using `window.location.href = "menteeUpcomingSession.php"`.

On the backend, the `book_session.php` script retrieves the session details to ensure it is still available, and if so, it proceeds to assign a `room_id` (generated using `uniqid()` if not already set). Figure 29 demonstrates this backend logic. The script then updates the session's status, assigns the current mentee, and saves these changes to the database using a prepared UPDATE statement. If the update is successful, a confirmation message including the generated `room_id` is returned, ensuring that the session is fully booked and ready for future use.

```

429 |
430 | function bookSession() {
431 |     if (!window.selectedSlotId) return;
432 |
433 |     fetch('book_session.php', {
434 |         method: 'POST',
435 |         headers: { 'Content-Type': 'application/x-www-form-urlencoded' },
436 |         body: 'session_id=' + window.selectedSlotId
437 |     }).then(response => response.text()).then(data => {
438 |         showModal("Your session has been successfully booked!");
439 |     });
440 | }
441 |

```

Figure 27 script for book session

```

442 |
443 | function showModal(message) {
444 |     document.getElementById("modalMessage").textContent = message;
445 |     document.getElementById("successModal").style.display = "flex";
446 | }
447 |
448 | function closeModal() {
449 |     window.location.href = "menteeUpcomingSession.php";
450 | }
451 |
452 |

```

Figure 28 Javascript to show pop-window

```

21 $sql = "SELECT mentor_id, date, time, room_id FROM sessions WHERE id = ? AND status = 'available'";
22 $stmt = $conn->prepare($sql);
23 $stmt->bind_param("i", $session_id);
24 $stmt->execute();
25 $result = $stmt->get_result();
26
27 if ($result->num_rows === 0) {
28     echo "X This session is no longer available!";
29     exit();
30 }
31
32 $session = $result->fetch_assoc();
33 $selected_date = $session['date'];
34 $selected_time = $session['time'];
35 $selected_mentor = $session['mentor_id'];
36 $room_id = $session['room_id'];
37 $stmt->close();
38
39 // إذا لم يكن موجود 'room_id' إنشاء
40 if (empty($room_id)) {
41     $room_id = uniqid('room_'); // جديد 'room_id' توليد
42 }
43
44 // تحديث الجلسة في قاعدة البيانات
45 $sql = "UPDATE sessions SET mentee_id = ?, status = 'booked', room_id = ? WHERE id = ?";
46 $stmt = $conn->prepare($sql);
47 $stmt->bind_param("isi", $mentee_id, $room_id, $session_id);
48
49 if ($stmt->execute()) {
50     session_write_close();
51     echo "✓ The session has been successfully booked! Room ID: " . htmlspecialchars($room_id);
52 } else {
53     echo "X Session not booked!";
54 }
55
56

```

Figure 29 book_sessions.php

6.2.3 ZEGOCLOUD API

ZEGOCLOUD is a real-time audio and video communication service that provides developers with powerful SDKs and APIs to build seamless video call, voice chat, and live streaming features into their applications. It offers low latency, high-quality media transmission, and a wide range of customizable UI components to suit different use cases from online meetings and virtual classrooms to one-on-one consultations. In this project, ZEGOCLOUD is used to power private video sessions between mentors and mentees, allowing for secure, interactive communication without the need to build a video infrastructure from scratch. By integrating ZEGOCLOUD, the platform ensures a smooth, reliable, and scalable video calling experience.

Figure 30 illustrates the process of initializing the video call session. The system first retrieves the roomID from the URL using the `$_GET` superglobal. This roomID uniquely identifies the virtual meeting room associated with a scheduled session. Using this identifier, the backend executes a prepared SQL statement to fetch the scheduled time and date of the session from the sessions table.

This ensures that the session details are securely retrieved and correspond to a valid booking. If the session is not found either due to an incorrect or expired roomID the system immediately halts execution and displays an error message, preventing unauthorized or invalid access to the video call environment. This step is essential for validating that the session exists and for calculating the correct timing logic later in the workflow.

```
<?php
2 session_start();
3 include 'db_connection.php';
4
5 if (!isset($_GET['roomID'])) {
6     die("Error : Room ID not found!");
7 }
8
9 $roomID = $_GET['roomID'];
10
11 // جلب وقت بدء الجلسة المجدول من قاعدة البيانات
12 $sql = "SELECT time, date FROM sessions WHERE room_id = ?";
13 $stmt = $conn->prepare($sql);
14 $stmt->bind_param("s", $roomID);
15 $stmt->execute();
16 $result = $stmt->get_result();
17 $session = $result->fetch_assoc();
18 $stmt->close();
19 $conn->close();
20
21 if (!$session) {
22     die("Error, the session not found");
23 }
24
```

Figure 30 retrieve Room ID

After confirming the session exists, the system sets the timezone to Asia/Riyadh using `date_default_timezone_set()` to ensure all time calculations are based on the correct local time. It then calculates the session's exact start time by combining the session's date and time fields and converting them into a Unix timestamp using `strtotime()`. To control the session's duration, the end time is set to exactly 45 minutes after the start by adding $45 * 60$ seconds. The current server time is also captured using `time()`, and the difference between the current time and the session's end time is computed to determine how many seconds remain. This remaining time is stored in the `$time_remaining` variable and later used in the JavaScript to automatically end the session when its duration expires. This mechanism ensures that sessions follow strict timing rules and helps maintain order and reliability across all scheduled video meetings.

```
25
26 date_default_timezone_set('Asia/Riyadh');
27 $session_start_time = strtotime($session['date'] . " " . $session['time']);
28 $session_end_time = $session_start_time + (45 * 60);
29 $current_time = time();
30 $time_remaining = max(0, $session_end_time - $current_time);
31
32 ?>
```

Figure 31 sets the timezone

Now the section of JavaScript plays a crucial role in enabling real-time video conferencing using the ZEGOCLOUD SDK and managing session timeouts automatically. It executes when the page finishes loading, as defined by the window.onload function.

1. Extracting the Room ID from the URL

As shown in Figure 32 the first step involves extracting the roomId parameter from the URL using a custom function getUrlParams(). This function splits the URL at the question mark (?) and parses the query string using URLSearchParams, returning a key-value object. The roomId is then retrieved from this object and stored in a constant. If no roomId is found, the script halts and alerts the user with an error.

```
52 <script>
53 window.onload = function () {
54     function getUrlParams(url) {
55         let urlStr = url.split('?')[1];
56         const urlSearchParams = new URLSearchParams(urlStr);
57         return Object.fromEntries(urlSearchParams.entries());
58     }
59
60     const roomId = getUrlParams(window.location.href)['roomId'];
61     if (!roomId) {
62         alert("Error : Room ID not found");
63         return;
64     }
65 }
```

Figure 32 Extracting the Room ID from the URL

2. Generating ZEGOCLOUD Access Token

Figure 33 shows how the script configures the session for the video call once the roomId is available. A unique user ID is randomly generated and combined with a base username (e.g., "user1234"). These values are used to generate a ZEGOCLOUD kitToken using the ZegoUIKitPrebuilt.generateKitTokenForTest() method, which requires:

- An appID (provided by ZEGOCLOUD),
- A serverSecret for authentication,
- The roomId where the call will take place,
- The generated userID and userName.

This token authenticates the user and grants them access to the video session.

```
65
66 const userID = Math.floor(Math.random() * 10000) + "";
67 const userName = "user" + userID;
68 const appID = 1039080135;
69 const serverSecret = "a503edcb519639c8aba065f3e67e6ba2";
70
71 const kitToken = ZegoUIKitPrebuilt.generateKitTokenForTest(appID, serverSecret, roomId, userID, userName);
72
```

Figure 33 Generating ZEGOCLOUD Access Token



3. Joining the Room and Configuring the Call Interface

A ZEGOCLOUD instance (zp) is created using `ZegoUIKitPrebuilt.create()` and initialized via `joinRoom()` with a set of configuration options tailored for private mentorship sessions.

The container property specifies the DOM element (`#root`) where the video call interface will be rendered.

The `sharedLinks` option enables participants to generate and share a unique link to the session room using the `roomId`.

The session is set to `VideoConference` mode, supporting real-time, interactive video calls between mentor and mentee.

Figure 34 illustrates the auto Activation of Devices :

- `turnOnMicrophoneWhenJoining: true`
- `turnOnCameraWhenJoining: true`

These options automatically enable the user's microphone and camera upon joining the session for a seamless start.

- `showAudioVideoSettingsButton: true`

Provides access to settings for selecting and testing audio/video devices without leaving the call.

- `showScreenSharingButton: true`

Enables mentors to share their screen it's useful for presenting slides, documents, or tools.

- `showTextChat: true`

Allows mentees and mentors to exchange messages, links, or notes in parallel with the video call.

- `showUserList: true`

Displays the list of connected participants, helping users confirm who is present in the session.

- `maxUsers: 2`

Restricts the room to only two participants (mentor and mentee) for focused, one-on-one engagement.

- `layout: "Auto"`
- `showLayoutButton: false`

The video layout is automatically adjusted based on screen size and video feeds, with no option for users to change it manually keeping the UI simple and distraction-free.

```

72 |
73 | const zp = ZegoUIKitPrebuilt.create(kitToken);
74 | zp.joinRoom({
75 |   container: document.querySelector("#root"),
76 |   sharedLinks: [{
77 |     name: 'Personal link',
78 |     url: window.location.protocol + '//' + window.location.host + window.location.pathname + '?roomId=' + roomId,
79 |   }],
80 |   scenario: {
81 |     mode: ZegoUIKitPrebuilt.VideoConference,
82 |   },
83 |   turnOnMicrophoneWhenJoining: true,
84 |   turnOnCameraWhenJoining: true,
85 |   showMyCameraToggleButton: true,
86 |   showMyMicrophoneToggleButton: true,
87 |   showAudioVideoSettingsButton: true,
88 |   showScreenSharingButton: true,
89 |   showTextChat: true,
90 |   showUserList: true,
91 |   maxUsers: 2,
92 |   layout: "Auto",
93 |   showLayoutButton: false,
94 | }

```

Figure 34 Joining the Room and Configuring the Call Interface

4. Time-Based Session Control

To ensure sessions are properly limited in duration, the script retrieves the remaining time from the backend PHP code (embedded with `<?php echo $time_remaining * 1000; ?>`) and stores it in the `timeRemaining` variable in milliseconds. This value represents the countdown until the session should end.

If `timeRemaining > 0`, a JavaScript `setTimeout` is set to trigger after the specified time elapses. When triggered:

- An alert notifies the user that the session has ended.

- A fetch() request is sent to update_session_status.php, passing the roomID to update the session's status in the database to 'completed'.
- If the backend confirms the session was updated successfully, the user is redirected to their respective past sessions page depending on their role.
- Any failure to update the session status is logged in the console for debugging.

If timeRemaining ≤ 0 , the session is already expired, and the user is immediately shown an alert and redirected to the appropriate fallback page.

```

106 let userRole = localStorage.getItem('userRole') || sessionStorage.getItem('userRole') || 'mentee';
107 console.log("User Role:", userRole);
108
109
110 let timeRemaining = <?php echo $time_remaining * 1000; ?> // **تحويله إلى ميلي ثانية**
111 console.log("Time remaining:", timeRemaining / 1000, "seconds");
112
113 // التأكد من أن الجلسة تنتهي في الوقت المحدد
114 if (timeRemaining > 0) {
115     setTimeout(function () {
116         alert("The session time has ended! You will be redirected shortly.");
117
118         // قاعدة البيانات "completed" لتحديث حالة الجلسة إلى
119         fetch('update_session_status.php?roomID=' + roomID)
120             .then(response => response.json())
121             .then(data => {
122                 console.log("Session Status Updated:", data);
123                 if (data.success) {
124                     if (userRole === "mentor") {
125                         window.location.href = "mentor_past_sessions.php"; // المentor ينتقل إلى صفحة إضافة جلسة جديدة
126                     } else {
127                         window.location.href = "past_sessions.php"; // Past Sessions المentee ينتقل إلى صفحة
128                     }
129                 } else {
130                     console.error("XError updating the session:", data.error);
131                 }
132             })
133             .catch(error => console.error('X Error while updating the session:', error));
134     }, timeRemaining); // على الوقت المتبقي سيتم إنهاء الجلسة بناءً
135 } else {
136     console.log("The session has already ended!");
137     alert("Session time is over!");
138     window.location.href = (userRole === "mentor") ? "mentor_past_sessions.php" : "past_sessions.php";
139 }
140
141

```

Figure 35 Time Remaining

6.2.4 Join Session Function

In the session joining system, both the mentor and mentee follow nearly identical logic for displaying and handling upcoming sessions. As shown in Figure 36 an SQL query retrieves sessions from the sessions table where the status = 'booked', filtered by either mentor_id or mentee_id depending on the logged-in user. A key part of the SQL is ORDER BY s.date DESC, s.time DESC, which ensures that the sessions are displayed with the most recent ones first.

```

8 $query = "SELECT s.id, s.date, s.time, s.room_id, m.mentor_id, m.profile_picture, u.first_name, u.last_name, f.field_name
9 FROM sessions s
10 JOIN mentors m ON s.mentor_id = m.mentor_id
11 JOIN users u ON m.user_id = u.user_id
12 LEFT JOIN field f ON m.field_id = f.id
13 WHERE s.mentee_id = ? AND s.status = 'booked'
14 ORDER BY s.date DESC, s.time DESC";
15
16
17 $stmt = $conn->prepare($query);
18 if (!$stmt) {
19     die("Error in preparing the query: " . $conn->error);
20 }
21
22 $stmt->bind_param('i', $mentee_id);
23 $stmt->execute();
24 $result = $stmt->get_result();
25
26

```

Figure 36 SQL query retrieves session

To determine whether a session can be joined, the system uses conditional checks based on the session's scheduled time and the current server time. If the session is scheduled for a future time (i.e., the session date is greater than the current date, or the session time is later on the same day), as shown in Figure 37 the system displays a disabled "Join" button that triggers a modal popup with the message "The session hasn't started yet. Please wait!". This is handled using the condition `if (\$session\_date > \$current\_date || (\$session\_date == \$current\_date && \$session\_time > \$current\_time))`.

As shown in Figure 38 if the session has already ended, determined by the condition $\$isEnded = (\$session_date < \$current_date \parallel (\$session_date == \$current_date \ \&\& \ \$session_end_timestamp < strtotime(\$current_time)))$, and the user did not join the session in time, the interface simply displays the message "Session Ended" in place of the join button. This ensures that users who miss their scheduled time are clearly informed that the session is no longer accessible.

To implement this logic accurately, the system first sets the timezone using `date\_default\_timezone\_set('Asia/Riyadh')` to ensure all comparisons are in local time. It then fetches the current time and date with `date("H:i:s")` and `date("Y-m-d")`. The session's original start time is converted using `strtotime(\$row['time'])`, and the session's end time is calculated by adding 45 minutes (45*60 seconds) to it. If the current time is greater than this calculated end time, the session is marked as ended.

Otherwise, if the session is currently active, the system displays an active "Join" button with an anchor tag linking to `WEB_UIKITS.php?roomID=...`, allowing the user to enter the video session.

```

252     if ($session_date > $current_date || ($session_date == $current_date && $session_time > $current_time)) {
253         echo "<button class='join-btn' onclick='showJoinModal(\"The session hasn't started yet. Please wait!\", \"wait\")'>Join</button>"
254     } elseif ($isEnded) {
255         echo "Session Ended";
256     }
257 } else {
258     echo "<a href='WEB_UIKITS.php?roomID=" . htmlspecialchars($room_id) . "' class='join-btn'>Join</a>";
259 }
260 }
261 }
262 }
263 }
264 }
265 }

```

Figure 37 The conditions in join

```

223 // حساب وقت الانتهاء
224 date_default_timezone_set('Asia/Riyadh');
225 $current_time = date("H:i:s");
226 $current_date = date("Y-m-d");
227
228 $session_time = date("H:i:s", strtotime($row['time']));
229 $session_date = $row['date'];
230 $room_id = $row['room_id'];
231
232 $session_end_timestamp = strtotime($session_time) + 45*60;
233 $isEnded = ($session_date < $current_date) || ($session_date == $current_date && $session_end_timestamp < strtotime($current_time));
234

```

Figure 38 end session

6.2.5 PHPMailer library

Within our platform, the PHPMailer library was used as a reliable and secure solution for sending emails. It was integrated using Composer and configured to send messages via SMTP, supporting various key user interactions across the system. [29]

One of the primary uses of PHPMailer is to send OTP (One-Time Password) verification codes during the registration process for both mentors and mentees. After entering their email address, the user receives a code via email which they must enter to complete their registration. This ensures that the email address is valid and helps maintain platform security.

PHPMailer is also used in the password reset process. When a user forgets their password, they can request a reset link, which is sent to their email. This link contains a secure token and redirects the user to the reset password page, where they can safely enter and set a new password.

In addition, PHPMailer is used to notify mentors about the status of their application. If a mentor is approved, they receive a confirmation email welcoming them to the platform. If the application is rejected, a polite rejection message is sent explaining that their request was not accepted at this time.



Through these features, PHPMailer plays a key role in improving user experience, strengthening platform security, and maintaining effective communication between the system and its users.

6.3 Implementation Challenges

Throughout the implementation process, we encountered several challenges and worked to overcome them, including:

- User Authentication and Email Verification:

One of the main challenges was building a secure sign-up system for both mentors and mentees, along with implementing OTP email verification for mentor and mentee registration. We used PHPMailer to send verification emails, and we faced some difficulties with its configuration such as email delays and deliverability issues. Ensuring a smooth the verification process without confusing the user required multiple tests and adjustments.

- User Interface Design and Usability:

Designing an interface that is both intuitive and functional for two different user types (mentors and mentees) posed several challenges. Balancing simplicity with features like session booking, ratings, and filters wasn't easy. We had to make careful design choices to keep things clear and easy to use without losing any important features.



- Rating System and Comment Handling:

Creating a flexible and dynamic rating system with comment support involved designing a UI that locks after submission, prevents duplicate ratings, and updates the mentor's average score in real time. Ensuring that this logic worked consistently across multiple pages required extra care in both frontend and backend development.

- Database Design and Interactions:

Designing the database to handle the relationships between users, sessions, and ratings was a challenge. We had to make sure the data stayed accurate and that everything worked smoothly when updating the interface in real time.

- Filtering and Search Functionality:

Implementing filtering and search features in the Mentor Center was more complicated than expected. We needed to allow mentees to easily find mentors based on specific fields and years of experience, without slowing down the platform. Making sure the filters worked accurately and updated results quickly required extra work on both the frontend and backend.

ASPIRA GitHub: https://github.com/nouraalw/2025_GP_09.git

ASPIRA Jira:

<https://atata4343.atlassian.net/jira/software/projects/SCRUM/boards/1?atlOrigin=eyJpIjoiOTVjZGFkYWQ4ODA0NGRiNWUwMWQ2YzNmM2UyZWYxODIiLCJwIjoiaj9>



chapter 7: **System testing**



7. System Testing

This chapter presents the system testing of ASPIRA platform, carried out by targeted users to evaluate its performance, usability, and overall effectiveness. It begins with User Acceptance Testing (UAT) to ensure the platform is both functional and user-friendly. The following subsection provides demographic details of the participants involved in the evaluation. Next, we showcase the questionnaire results. The chapter concludes with a discussion analyzing the results to highlight ASPIRA's strengths and suggest areas for enhancement.

7.1 User Acceptance Testing

User Acceptance Testing (UAT) is a system evaluation that verifies whether the developed platform meets business requirements and is ready for real-world use. It emphasizes evaluating the system from the perspective of actual users, focusing on interface usability, technical performance, and identifying both strengths and areas for enhancement. For this evaluation, testing was carried out through the platform's web interface by a group of selected end users specifically, three mentees and two mentors and one admin. Participants were asked to perform key tasks within the platform to simulate real usage scenarios.

Mentor's tasks:

- Register
- Log in
- Log out
- Reset password
- Add session
- View available and Upcoming sessions
- Join session
- View past sessions



Mentee's tasks:

- Register
- Log in
- Log out
- Reset password
- View Mentor center
- Search mentors by name
- Filter mentors by field and years of experience
- View ratings and feedback submitted by mentees for the mentor
- Book private session
- View upcoming sessions
- Join session
- rate mentor and put comment
- View past sessions

Admin tasks:

- Log in
- Log out
- Accept/Reject mentor requests

In addition to completing these tasks, they also provided feedback by filling out a detailed questionnaire (Appendix C and Appendix D) designed to capture their overall experience, satisfaction, and suggestions for improvement.

7.1.1 Demographics of Participants

To better understand the background of the individuals involved in the User Acceptance Testing (UAT), this section presents the demographic information of the selected testers. Gathering these details helps provide context for interpreting the feedback and evaluating how well the platform meets the needs of its target users. The participants include mentors, mentees and admin, and their demographic data is summarized in Table [9] below.

Table 9 Demographics of Participants

	Age	Gender	Role	level of education	Technological skills
Participant 1	23 - 49	Female	Mentor	Master's degree	intermediate
Participant 2	18 - 22	Male	Mentee	Bachelor's degree	Advance
Participant 3	23 - 49	Female	Mentee	Bachelor's degree	Advance
Participant 4	23 - 49	Male	Mentor	Master's degree	Advance
Participant 5	18 - 22	Female	Mentee	Bachelor's degree	intermediate
Participant 6	23 - 49	Male	Admin	Master's degree	Advance

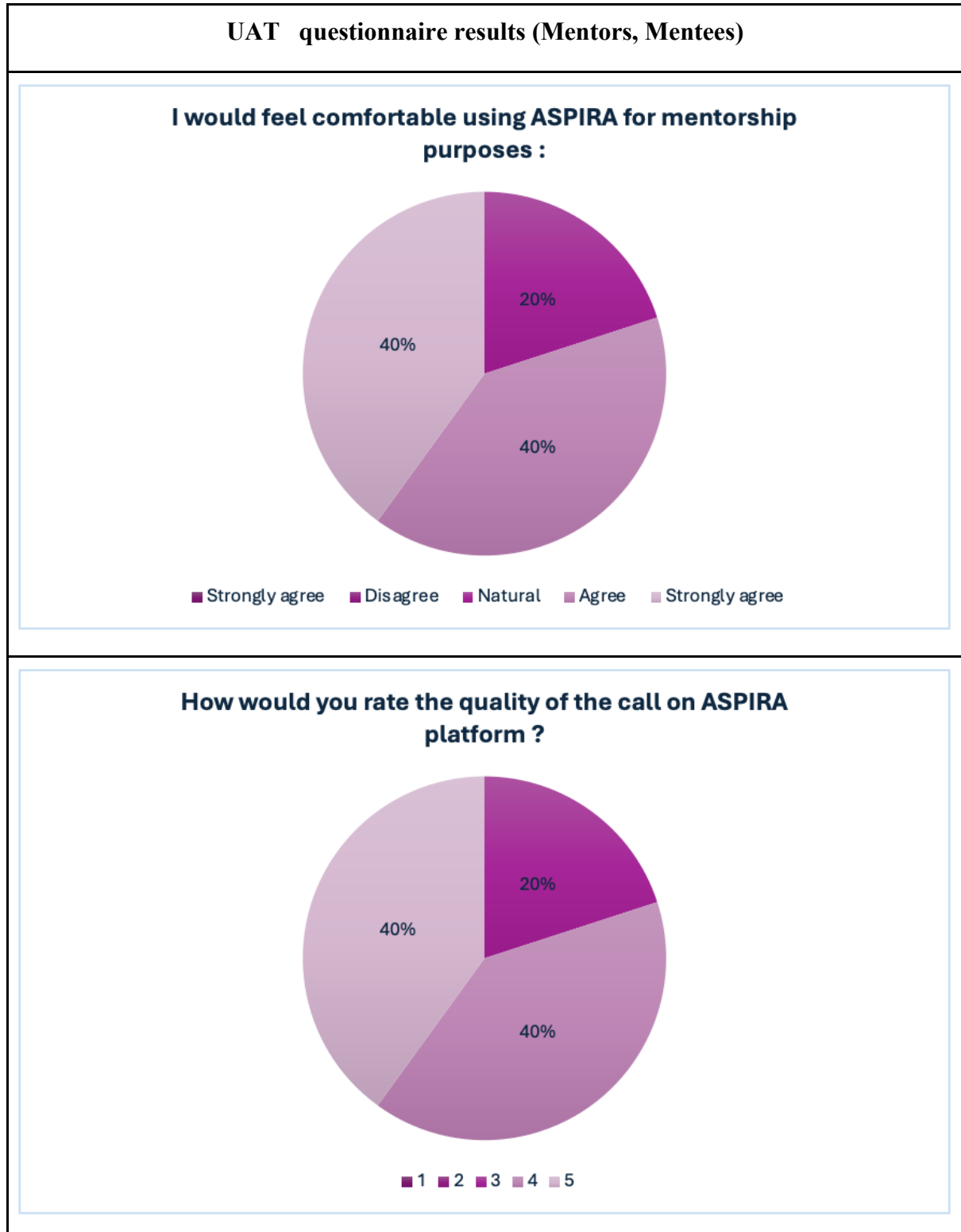
7.1.2 Questionnaire/Interview Results

This section presents the feedback collected from mentors, mentees and admin during the User Acceptance Testing (UAT), based on the questionnaire (Appendix C and Appendix D) as outlined in Table [10] and Table [11] below. A score of 1 indicates low, and 5 indicates high.

Based on the mentors and mentees questionnaire responses, most users found the platform easy to use 4 out of 5 said it was user-friendly, and one strongly agreed. Regarding the platform's visual design, 3 agreed it was clear and appealing, 1 strongly agreed, and 1 was neutral. For call quality, 2 participants rated it a 5, 2 rated it a 4, and 1 rated it a 3. When asked about using ASPIRA platform regularly for mentorship purposes 2 strongly agreed, 2 agreed, and 1 was neutral. Three users had no issues, while two reported a slight delay when joining

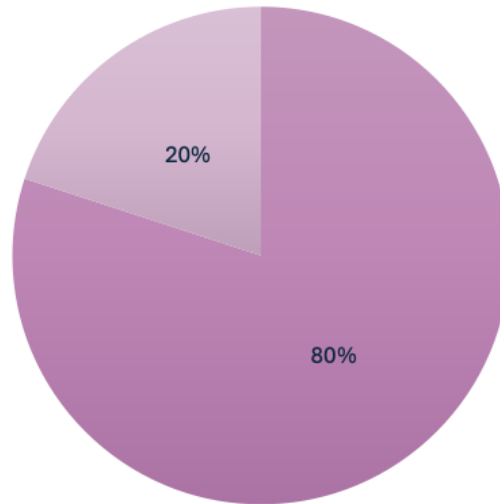
the call. One user suggested adding Arabic language support, and all 5 participants said they would recommend ASPIRA to others.

Table 10 UAT questionnaire results (Mentors, Mentees)



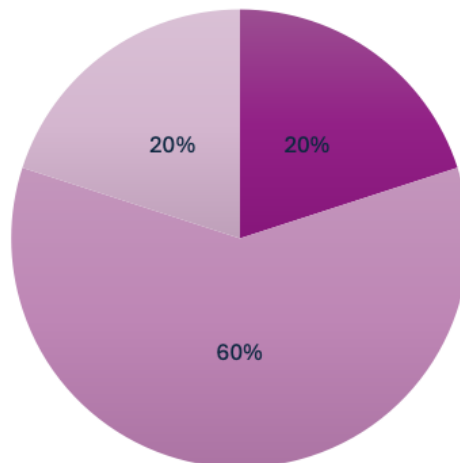
UAT questionnaire results (Mentors, Mentees)

The platform was easy to navigate and user-friendly :



Strongly agree Disagree Natural Agree Strongly agree

The layout and design of the platform are clear and visually appealing :



Strongly agree Disagree Natural Agree Strongly agree



UAT questionnaire results (Mentors, Mentees)

What issues or difficulties did you encounter, if any?

5 responses

The call took a few seconds to connect

i didn't encounter any difficulties

no

none

I faced a slight delay when joining the session but overall the experience was smooth

Do you have any comments or suggestions for improving ASPIRA ?

5 responses

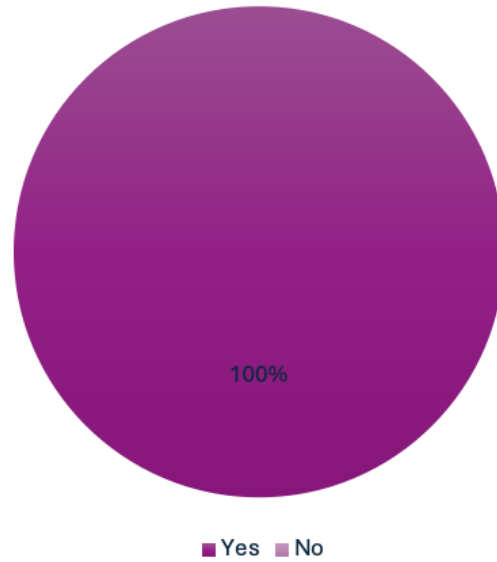
no

add Arabic language to the platform

great platform

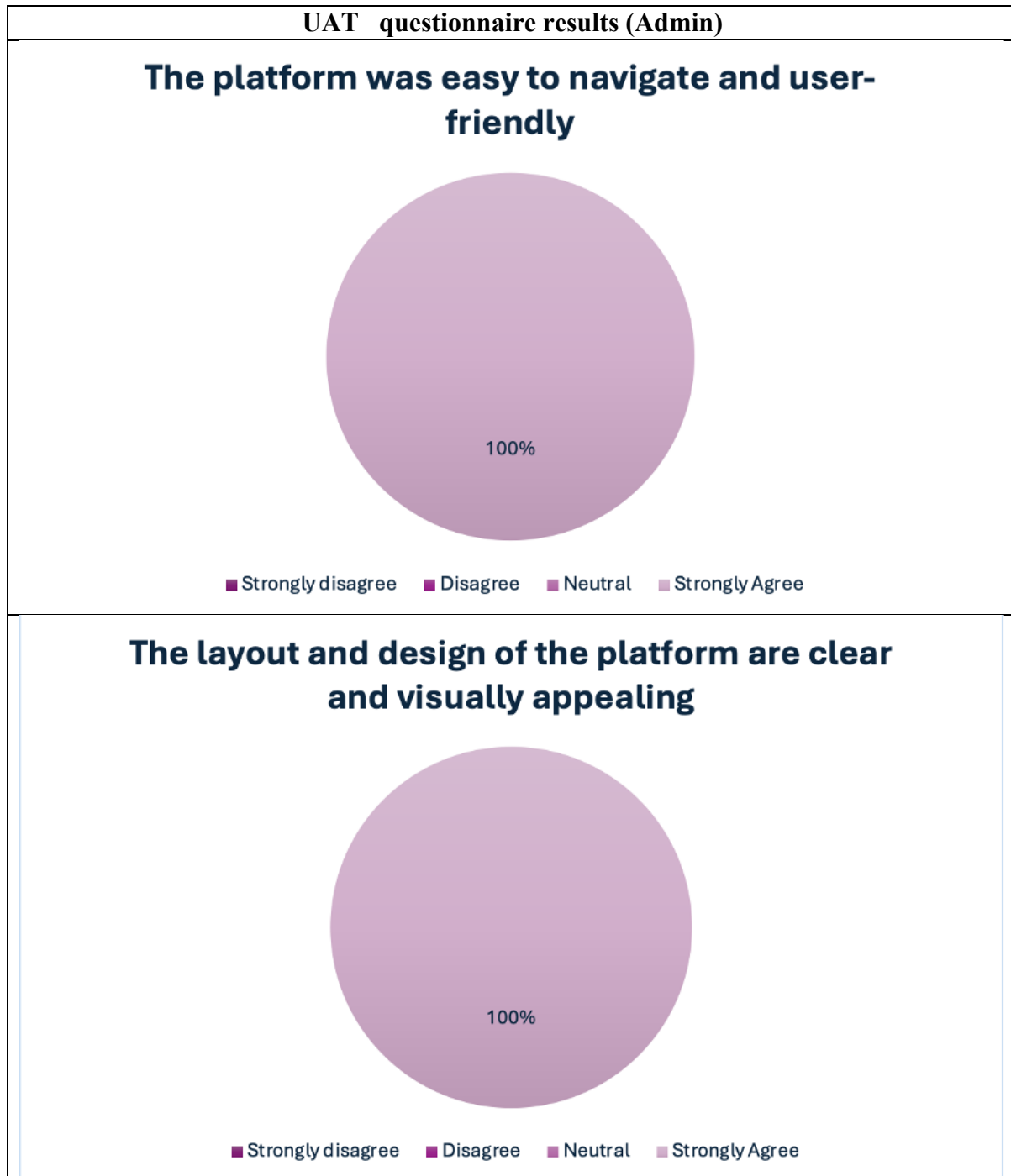
UAT questionnaire results (Mentors, Mentees)

Would you recommend ASPIRA ?



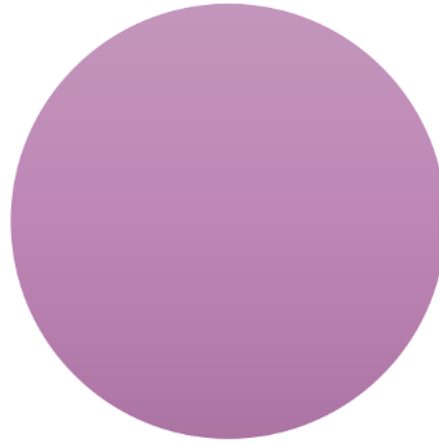
During the User Acceptance Testing for the Admin role, the participant was asked several questions regarding his experience. When asked if the platform was easy to navigate and user-friendly, the participant strongly agreed as shown below. In addition, the clarity and visual appeal of the platform's layout and design, the participant again strongly agreed, when asked whether it was simple to view mentor profiles and attached documents, the participant answered "Yes". In response to whether any issues or difficulties were encountered during the testing process, the user responded "No." Additionally, when asked if they had any comments or suggestions for improving the platform, the participant commented, "No, the platform was easy to use." Overall, the response confirmed that the user found the system intuitive, visually clear, and efficient for managing mentor applications.

Table 11 UAT questionnaire results (Admin)





Did you find it simple to view mentor profile and attached documents ?



■ No ■ Yes

What issues or difficulties did you encounter, if any?

1 response

No

Do you have any comments or suggestions for improving ASPIRA ?

1 response

No the platform was easy to use

7.2 Discussion

The User Acceptance Testing (UAT) results reveal highly positive feedback from mentors, mentees and the admin participant, indicating that the ASPIRA platform effectively meets user needs and business requirements.

The majority of users reported a smooth and satisfying experience. Notably, all mentors and mentee stated they would recommend ASPIRA to others, highlighting a high level of trust and perceived value in the platform. Most users found the platform easy to navigate and user-friendly, with 4 participants agreeing and 1 along with the admin participant strongly agreeing with this statement. In terms of design, 3 agreed and 2 strongly agreed, while one participant chose natural, that the platform's layout is clear and visually appealing, indicating that minor visual improvements could enhance the experience further.

Additionally, the Admin participant confirmed that it was simple to view mentor profiles and their attached documents, such as CV and certificates, indicating that the system effectively supports administrative review tasks

Feedback on the statement "I would feel comfortable using ASPIRA regularly for mentorship purposes" reflected strong user confidence in the platform. Four out of five of mentors and mentees participants agreed or strongly agreed, while one responded neutrally. No users expressed discomfort or disagreement, reinforcing the platform's suitability for continuous and real-world use.

Regarding technical functionality, 4 of the users rated the call quality 4 or 5 out of 5, showing that the system performs well in real-time interactions, one user gave a rating of 3. Out of five respondents, three explicitly stated they experienced no difficulties. However, two users mentioned minor problems related to session connectivity specifically, a slight delay before the video call began. These delays did not disrupt the overall session's flow.

User feedback, comments and suggestions were minimal but constructive. One tester suggested adding Arabic language support, which would increase accessibility for a wider audience, therefore, providing Arabic language will be part of our future work. Another participant described ASPIRA as a "great platform," and the admin participant stated "No the platform was easy to use" while some had no suggestions implying general satisfaction.



In conclusion the User Acceptance Testing results indicated that the ASPIRA platform successfully meets the expectations of its target users, offering a reliable, user-friendly, and effective mentorship experience. Testers responded positively across all key areas, from navigation and call quality to overall satisfaction and platform usability.



chapter 8: Conclusions



8. Conclusions and Future Work

The integration of artificial intelligence within mentorship and career development reaches its greatest milestone through ASPIRA. ASPIRA was established with Saudi Arabia's youth support in mind to provide an organized scalable platform that links mentees with experienced mentors while supplying necessary tools for professional growth.

We built the system to respond to genuine academic-to-workplace transition problems faced by students and recent graduates. Career-based problems that youth face are their restricted access to mentorship services combined with uncertain career paths and weak skills training materials. The platform of ASPIRA includes multiple features which combine one-on-one mentorship sessions with group mentorship sessions while delivering AI-based educational recommendations to users in addition to providing an automatic CV builder and an intelligent chatbot for instant help.

The development work combined different fields of expertise which included user research together with requirement engineering along with literature reviews and architectural design and advanced software development methods. Each aspect of the platform development depended on user feedback to create a practical solution which continues providing relevant and usable features. Multiple rounds of development along with strategic plans and joint work efforts transformed ASPIRA into its current status as a modern digital mentorship platform meeting today's educational and professional criteria.

8.1 Global and local impact

The local impact of ASPIRA in Saudi Arabia is profound. The platform directly aligns with Vision 2030 by enabling Saudi youth to become job-market ready, develop soft and hard skills, and access meaningful mentorship without geographical barriers. ASPIRA fosters a supportive ecosystem where mentees gain confidence, explore new career paths, and build personal development strategies with the help of mentors who bring industry expertise.

Through the AI-powered features, ASPIRA not only increases access to mentorship but also personalizes the experience, ensuring each mentee is matched with the most suitable mentor



based on their interests and goals. This leads to more effective outcomes, higher satisfaction, and better preparedness for professional life.

On a global scale, ASPIRA has the potential to serve as a model for other countries facing similar gaps in career support systems. By combining mentorship with AI technologies like recommendation systems and chatbots, ASPIRA introduces a scalable framework that can be localized for different regions. It serves as an example of how technology can democratize access to professional guidance, enabling students and early-career professionals to succeed, regardless of background or resources.

Furthermore, ASPIRA contributes to the global conversation on digital education and workforce readiness, offering insights into how technology can support lifelong learning, personalized development, and skill alignment with evolving job markets.

8.2 Problems and challenges encountered during the software development

Throughout the software development life cycle, the ASPIRA team encountered various technical and operational challenges, which required agile problem-solving and continuous learning. These challenges included:

1. API Integration and Time Constraints

During development, a key external API unexpectedly became unavailable, disrupting platform progress. The team had to invest time in finding a reliable alternative and updating the integration logic, which introduced delays and added technical complexity.

2. Mentor and Mentee Role Separation

A critical bug was identified where booked sessions incorrectly recorded the mentee ID as the mentor ID. Fixing this required restructuring backend logic and updating database queries to ensure role integrity.



3. Rating System Integration

Implementing a one-time, non-editable 5-star rating system with comment support demanded careful design. Preventing duplicate submissions and dynamically updating mentor averages required advanced state handling.

4. User Experience Consistency

Creating a consistent, intuitive interface for mentors, mentees, and admins required multiple iterations. Balancing feature complexity with ease of use was a key design challenge.

5. Database Optimization

As features grew, so did the complexity of database interactions. The team worked on optimizing SQL queries, preventing redundancy, and ensuring referential integrity across tables such as users, sessions, ratings.

8.3 Limitations of the system

As with any project, ASPIRA has a number of limitations, many of which have been identified for future enhancement:

- Language Support

Currently, ASPIRA supports English only, which may exclude Arabic-speaking users or non-English speakers from fully utilizing the platform.



- API Call Limitations

ASPIRA's video session feature relies on a third-party API that provides a limited number of free calling minutes per month. This restriction could impact session availability or quality if usage exceeds the quota, especially during periods of high activity.

- Mobile Responsiveness

While the platform is accessible via browser, full optimization for mobile devices is still under development, potentially affecting the experience on small screens.

- Lack of In-App Messaging

Communication between mentors and mentees is limited to session time. There is no real-time chat feature to enable discussions outside scheduled meetings.

8.4 The main contribution of the project

ASPIRA offers several distinct contributions to the field of educational technology and career development:

1. End-to-End Mentorship Platform

- ASPIRA offers a full ecosystem for career guidance, from mentor discovery and booking to session summaries, feedback, and real-time communication.



2. AI-Powered Chatbot

- The integration of a GPT-4-based chatbot offers mentees real-time, context-aware responses to common career and platform-related questions, increasing accessibility and user engagement.

3. Course Recommendation Engine

- Using a content-based filtering approach, the system suggests tailored learning materials that support mentees' career paths, ensuring continuous growth outside of mentorship sessions.

4. CV Builder

- A guided CV builder to help mentees create professional resumes with ease, supporting their job applications and career development.

8.5 Future work

To enhance ASPIRA and expand its impact, the team has identified the following areas for future development:

1. Arabic Language Support

- Translate and localize the entire platform interface and chatbot to support bilingual users in Saudi Arabia and other Arabic-speaking countries.



2. Mobile Application Development

- Launch dedicated iOS and Android apps with optimized interfaces for mobile use, ensuring greater accessibility and smoother user experience.

3. Real-Time Messaging System

- Add secure, instant chat functionality between mentors and mentees to allow ongoing communication, file sharing, and scheduling support.

4. Calendar Integration and Session Reminders

- Sync booked sessions with Google Calendar, Apple Calendar, or Outlook, with automatic reminders via email or push notification.

5. Speech-Based Chatbot Support

- Enhance the chatbot to include voice interaction features, especially useful for users with disabilities or those preferring hands-free usage.

6. Automated Mentor Onboarding

- Implement AI-powered CV parsing and scoring to assist admins in evaluating mentor applications and reduce manual workload.



chapter 9: References



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10. Appendix A : interviews

Table 12 Interview with mentor [1]

Interview with mentor [1]	
Interviewee: Galah alharbi	Interviewer: Aeshah Almukhlifi
Location: phone call	Date: 20 Jan 2025
Question:	Answers and Notes:
Q1: Can you tell me about your major, your experience and what inspired you to start mentoring ?	I have over 10 years of experience in software development, and I love my job. I started mentoring students in college, helping younger students, and I found it fun. Since then, I've continued mentoring because I enjoy helping others.
Q2: What strategies do you use to help mentees overcome challenges or achieve their goals?	I listen to what the mentee needs help with and what they want to achieve. Then, we break it into smaller steps and work on them one by one. I share advice from my experience and check in to support them.
Q3: What do you find most challenging when communicating with mentees?	The most challenging part is understanding what mentees really need, especially when they text me at an unsuitable time and seem angry. It can be hard to address their concerns effectively in those moments, so I try to stay calm, understand their frustration, and respond clearly once things settle.
Q4:How do you think technology can help improve the mentoring process?	I see the technology makes mentoring easier by helping me stay organized and share resources quickly. It also makes it simple to connect and communicate anytime, which is really helpful.



Q5: Do you believe a platform offering flexibility in communication methods (like video) would make the mentoring process more effective?

Yes, I think having options like video would make mentoring better. Video calls help with clearer communication and understanding emotions, which can be harder through text.

Table 13 Interview with mentor [2]

Interview with mentor [2]	
Interviewee name : Shaden Alshehri	Interviewer : Ghadi Alzahrani
Location: Zoom meeting	Date: 20 Jan 2025
Question:	Answers and Notes:
Q1: Can you tell me about your major, your experience and what inspired you to start mentoring ?	My major is business administration, and I've been working in marketing for 8 years. I'm specialized in branding. I've always enjoyed sharing my knowledge and helping others.
Q2: What strategies do you use to help mentees overcome challenges or achieve their goals?	I focus on helping mentees understand their goals clearly and then work with them to create a step-by-step plan. on the other hand for the challenges first listen to their challenges and understand what's holding them back. Then, I work with them to create a clear plan and provide resources that can help.
Q3: What do you find most challenging when communicating with mentees?	For me, the hardest part is when mentees don't communicate regularly or delay responses. It makes it difficult to stay aligned with their progress.



Q4:How do you think technology can help improve the mentoring process?	can make mentoring more accessible and efficient. and allow mentors and mentees to collaborate easily, even if they're in different locations.
Q5:Do you believe a platform offering flexibility in communication methods (like video) would make the mentoring process more effective?	Of course. Video communication allows for better interaction, as you can see expressions and body language, making the mentoring process clearer, engaging and real.

Table 14 Interview with mentor [3]

Interview with mentor [3]	
Interviewee name : Haifa Alsaif	Interviewer : Ghadi Alzahrani
Location: Zoom meeting	Date: 20 Jan 2025
Question:	Answers and Notes:
Q1: Can you tell me about your major, your experience and what inspired you to start mentoring ?	I studied Finance and investments and have been working in project management since graduation for almost 6 years. I started mentoring when I noticed how many people struggled with planning. I wanted to share the strategies that worked for me to help others.
Q2: What strategies do you use to help mentees overcome challenges or achieve their goals?	My strategy is to identify the root cause of any challenge. I then guide them in setting realistic goals and finding practical solutions. I like to use real world examples from my own experience to show them how to handle similar situations.
Q3: What do you find most challenging when communicating with mentees?	The most challenging part is when mentees are not clear about what they want to achieve or struggle to explain their challenges.
Q4:How do you think technology can help improve the mentoring process?	Technology makes mentoring easier by helping with things like scheduling meetings and staying in touch. It makes communication faster and keeps everything simple.



Q5:Do you believe a platform offering flexibility in communication methods (like video) would make the mentoring process more effective?

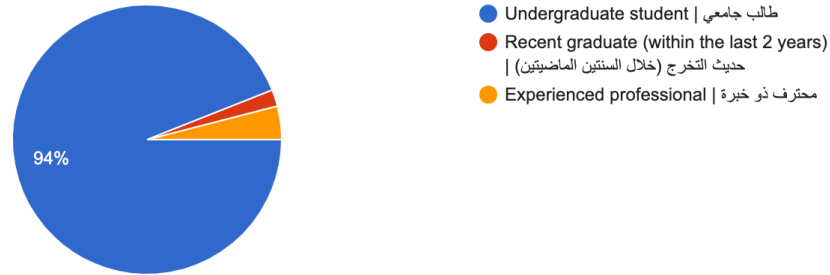
Yes, it would definitely improve mentoring. It makes conversations more personal and helps in explaining things more effectively.

11. Appendix B : questionnaire

1- What is your current status? | ما هو وضعك الحالي؟

[Copy chart](#)

50 responses

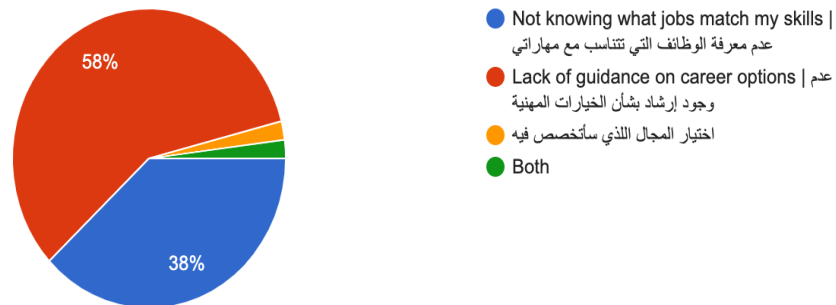


2- What are the biggest challenges you face when deciding on a career path ? | ما هي أكبر التحديات التي تواجهك عند اتخاذ قرار بشأن المسار الوظيفي؟

[Copy chart](#)

هي أكبر التحديات التي تواجهك عند اتخاذ قرار بشأن المسار الوظيفي؟

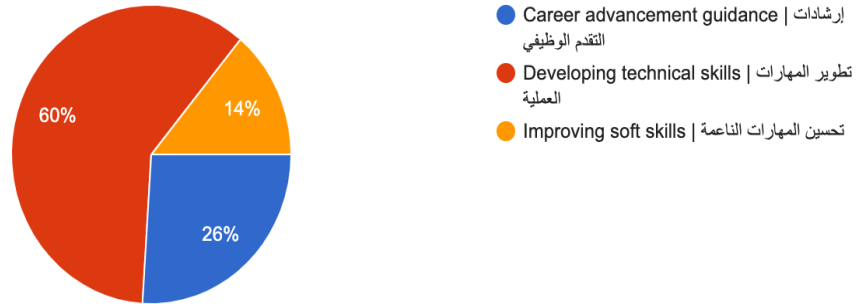
50 responses



3- Which mentoring topics do you consider most important for your professional growth? | ما هي مواضيع الإرشاد التي تعتبرها الأكثر أهمية لنموك المهني؟

[Copy chart](#)

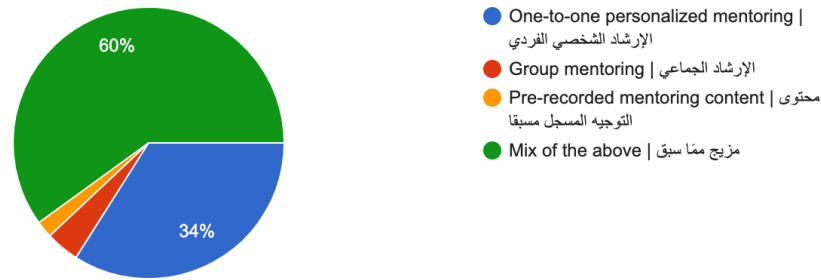
50 responses



4- What is your preferred way of receiving career guidance? | ما هي الطريقة المفضلة لديك لتلقي الإرشاد المهني؟

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50 responses

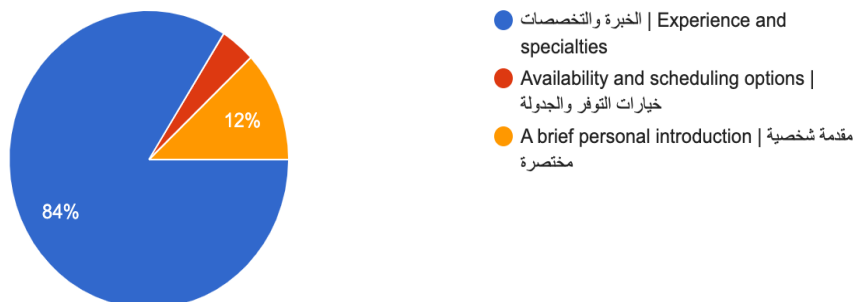


5- If you had the opportunity to connect with a mentor, what would catch your attention the most in their profile? | إذا أتاحت لك الفرصة للتواصل مع أحد المرشدين، ما هو أكثر ما يلفت انتباهك في ملفه الشخصي؟

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ما يلفت انتباهك في ملفه الشخصي؟

50 responses

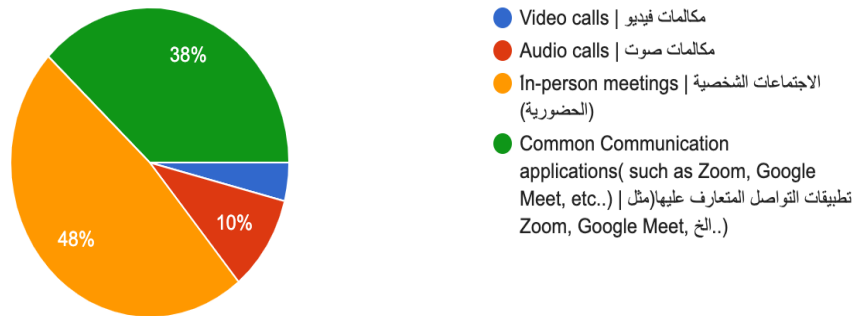


6- What is your preferred mode of Communication with Mentors? | ماهي وسيلة التواصل المفضلة لديك مع المرشدين؟

التواصل المفضلة لديك مع المرشدين؟

50 responses

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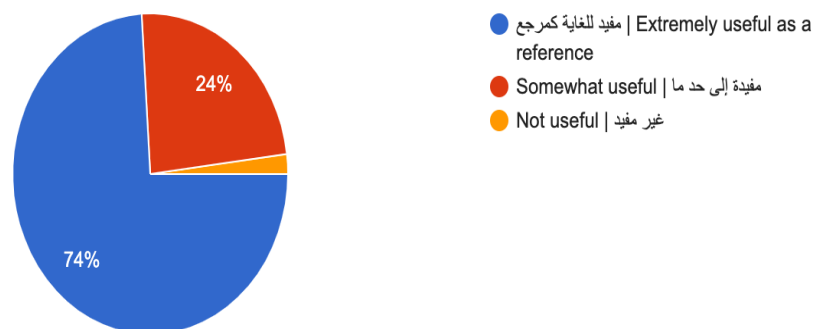


7- How useful would you find a history of your previous sessions? | ما مدى فائدة وجود مرجع لجلساتك السابقة؟

وجود مرجع لجلساتك السابقة؟

50 responses

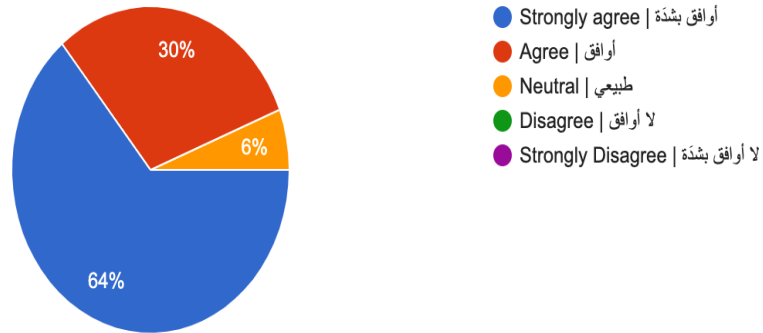
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8- Having a rating feature for mentors would make me more confident in selecting the right mentor. | وجود خاصية التقييم للمرشدين سيجعلني أكثر ثقة في اختيار المرشد المناسب.

50 responses

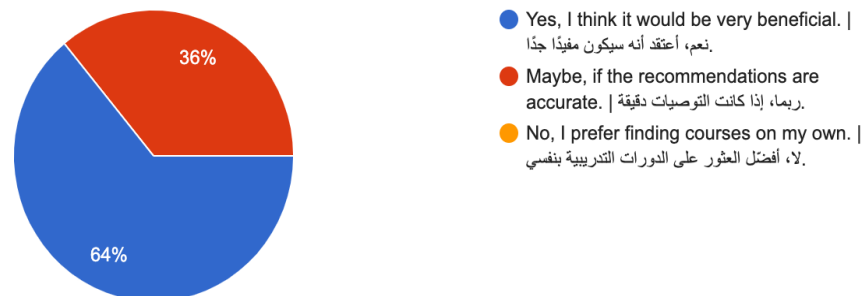
[Copy chart](#)



9- Would you be interested in a recommendation system that suggests courses tailored to your interests and career goals? | هل أنت مهتم بنظام التوصيات الذي يقترح دورات مصممة خصيصًا لتناسب اهتماماتك وأهدافك المهنية؟

50 responses

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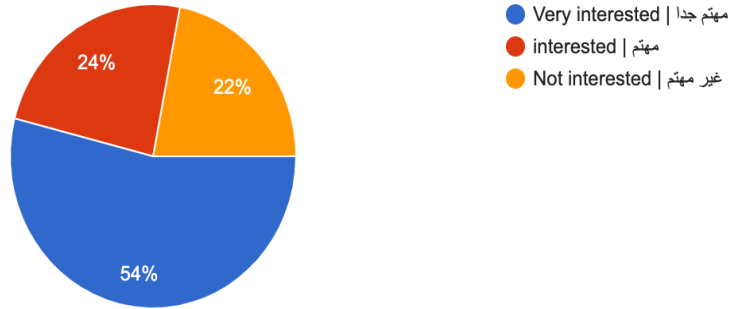




10- How interested are you in having an AI-powered chatbot as part of your mentorship experience? | ما مدى اهتمامك بتوافر روبوت محادثة يعمل بالذكاء الاصطناعي كجزء من تجربتك الإرشادية؟

50 responses

[Copy chart](#)



13. Appendix D : UAT questionnaire (Admin)

ASPIRA
User acceptance testing questionnaire (Admin)

ghassid@gmail.com [Switch account](#)

ES Not shared

* indicates required question

Age *

☐ 18 - 23

☐ 23 - 49

☐ 50 and above

What is your gender ? *

☐ Female

☐ Male

What is your role ? *

☐ Mentor

☐ Mentee

☐ Admin

What is your highest level of education? *

☐ High school

☐ Diploma

☐ Bachelor's degree

☐ Master's degree

☐ PhD

Technological skills : *

☐ Novice

☐ Intermediate

☐ Advanced

The platform was easy to navigate and user-friendly *

☐ Strongly disagree

☐ Disagree

☐ Neutral

☐ Agree

☐ Strongly Agree

The layout and design of the platform are clear and visually appealing *

☐ Strongly disagree

☐ Disagree

☐ Neutral

☐ Agree

☐ Strongly Agree

Did you find it simple to view mentor profile and attached documents ? *

☐ Yes

☐ No

What issues or difficulties did you encounter, if any? *

Your answer

Do you have any comments or suggestions for improving ASPIRA ? *

Your answer

[Submit](#) [Clear form](#)

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